B.Tech. BCSE497J - Project-I

HYBRID CALIBRATION FRAMEWORK FOR BRAIN TUMOR DETECTION

Submitted in partial fulfillment of the requirements for the degree of

Bachelor of Technology

in

Computer Science and Engineering

By

22BCE3317 AISHWARYA S

22BCE3439 DEVESH CHAHAR

22BCE3451 PRANAV AGRAWAL

Under the Supervision of

DR. BIJI C.L

Associate Professor Grade 1

School of Computer Science and Engineering (SCOPE)



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DECLARATION

I hereby declare that the project entitled **"Hybrid Calibration Framework for Brain Tumor Detection"** submitted by me, for the award of the degree of *Bachelor of Technology in Computer Science and Engineering* to VIT is a record of bonafide work carried out by me under the supervision of **Dr. Biji C.L.**

I further declare that the work reported in this project has not been submitted and will not be submitted, either in part or in full, for the award of any other degree or diploma in this institute or any other institute or university.

Place : Vellore

Date :

Signature of the Candidate

CERTIFICATE

This is to certify that the project entitled **"Hybrid Calibration Framework for Brain Tumor Detection"** submitted by **Aishwarya S (22BCE3317), Devesh Chahar (22BCE3439), Pranav Agrawal (22BCE3451), School of Computer Science and Engineering**, VIT, for the award of the degree of *Bachelor of Technology in Computer Science and Engineering*, is a record of bonafide work carried out by him / her under my supervision during Fall Semester 2025-2026, as per the VIT code of academic and research ethics.

The contents of this report have not been submitted and will not be submitted either in part or in full, for the award of any other degree or diploma in this institute orany other institute or university. The project fulfills the requirements and regulations ofthe University and in my opinion meets the necessary standards for submission.

Place : Vellore

Date :

Signature of the Guide

Internal Examiner **External Examiner**

Dr. Boominathan P

HOD – Department of Software Systems, SCOPE

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I am immensely thankful to my project guide, **Dr. Biji C.L.**, for her dedicated mentorship and invaluable feedback. Her patience, knowledge, and encouragement have been pivotal in the successful completion of this project. My supervisor's willingness to share her expertise and provide thoughtful guidance has been instrumental in refining my ideas and methodologies. Her support has not only contributed to the success of this project but has also enriched my overall academic experience.

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List of Abbreviations

2G	Second Generation
3GPP	Third Generation Partnership Project
3G	Third Generation
4G	Fourth Generation
AWGN	Additive White Gaussian Noise
BBE	Background Block Error
CSMA	Carrier Sense Multiple Access
DAS	Direct Attached Storage

Note: All the abbreviations should follow Alphabetical order

Symbols and Notations

δf	CFO
ε	NCFO

ABSTRACT

The diagnosis of brain tumors from Magnetic Resonance Imaging (MRI) is a critical, high-stakes task where clinical decisions depend on accuracy and reliability. While modern deep learning models like YOLOv8 have achieved state-of-the-art accuracy in detecting tumors, their practical adoption is hindered by a significant 'trust gap'. These models are often mis calibrated, displaying a high degree of overconfidence that does not align with their real-world performance. Our baseline analysis quantified this problem, revealing that a high-accuracy YOLOv8s model exhibited a substantial Expected Calibration Error (ECE) of 0.192 on our test set. This mismatch between reported confidence and actual correctness makes the model unsafe for automated clinical triage.

This project directly confronts this challenge by engineering and validating a novel, Clinically-Aware Meta-Model Calibration Framework. Instead of applying a single, generic correction factor, our approach trains a second-opinion classifier (a Gradient Boosting model) that learns to predict the true probability of correctness for any given detection. This meta-model is 'clinically-aware' because it was trained to evaluate a detection using a rich vector of contextual features that a radiologist would use: the baseline model's raw confidence (logit), the objective image quality (using the no-reference BRISQUE score), and key spatial features (bounding box size and location).

This framework was rigorously developed by training the meta-model on a dedicated validation dataset. We then evaluated it on a completely unseen test set to ensure an unbiased measure of performance. The results were definitive: our BRISQUE-enhanced meta-model reduced the ECE from 0.192 to 0.044, a 77.2% reduction in calibration error. This superior reliability was further confirmed by a 28.2% improvement in the Brier score.

To demonstrate the real-world utility of this work, we integrated our trained meta-model into an end-to-end Intelligent Triage Application. This tool uses the meta-model's now-trustworthy probabilities to automatically sort cases into three distinct clinical queues. High-confidence tumor detections are flagged for the 'Red Queue (Urgent Review)'. High-quality, high-confidence normal scans are placed in the 'Green Queue (Routine Review)', safely reducing clinician workload. Critically, all ambiguous, low-confidence, or low-quality scans are caught and placed in the 'Yellow Queue (Standard Review)', creating a vital safety net. This project successfully demonstrates the complete pipeline from identifying a core reliability flaw in AI to building a robust, context-aware, and deployable tool that bridges the trust gap for clinical use.

CHAPTER 1

INTRODUCTION

1.1 BACKGROUND

Among the most severe and life-altering neurological conditions, a brain tumor diagnosis is a critical event. These abnormal growths, arising in the brain or its surrounding tissues, can lead to devastating complications, from cognitive decline and seizures to, in many cases, mortality if not caught and treated in time. For this reason, early and, crucially, accurate diagnosis is the single most important factor in guiding a patient's treatment plan and improving their chances of a positive outcome.

For decades, the cornerstone of this diagnostic process has been Magnetic Resonance Imaging (MRI). Unlike other methods, MRI offers an unparalleled, high-resolution view into the soft tissues of the brain, allowing clinicians to visualize tumor structures in detail without using ionizing radiation. But this gold standard is not without its human challenges. The sheer volume of diagnostic scans is increasing, placing an immense workload on radiologists. Interpreting these complex images is a demanding task, subject to human subjectivity and fatigue, especially when faced with the wide variability in tumor shapes, sizes, and compositions.

This is where Artificial Intelligence, specifically deep learning, has shown incredible promise. In the last ten years, Convolutional Neural Networks (CNNs) have revolutionized medical image analysis. We have progressed from models that could simply classify an image to advanced, real-time object detectors like the YOLO (You Only Look Once) family, which can pinpoint the exact location of a tumor within seconds.

However, this impressive technological leap masks a deep and dangerous flaw. These models, while highly accurate, are fundamentally unreliable. They often exhibit extreme overconfidence, reporting a ‘95% confidence’ on a prediction that is, in reality, far less certain. This disconnect between a model's stated confidence and the actual likelihood of it being correct is known as miscalibration. In a high-stakes field like medicine, this ‘trust gap’ is not just a statistical curiosity; it is a critical barrier to safe, real-world adoption.

1.2 MOTIVATION

While countless research papers boast near-perfect accuracy, very few of these AI tools are actually used in hospitals. The reason is simple: a doctor cannot trust a ‘black box’. A model's prediction is useless, and potentially dangerous, if the confidence score attached to it is meaningless.

This is the ‘trust gap’ that motivates our entire project. Imagine an AI that correctly identifies a scan as ‘no-tumor’ with 90% confidence. Can a radiologist safely move this case to the bottom of the pile? What if that 90% is really only 60% accurate? Conversely, imagine a model detects a tiny tumor with only 70% confidence. A busy clinician might dismiss this as a low-confidence error. But what if the model was correct, just uncertain due to a blurry scan?

For an AI to be a true partner to a clinician, it must be more than just accurate. It must be reliable. Its confidence must be meaningful. A doctor needs to know that when the model says it is 99% sure, it is actually 99% sure.

Our motivation, therefore, is not to build a model that is 1% more accurate. It is to build a model whose confidence can be trusted. We aim to create a framework that transforms a powerful but unreliable AI into a dependable, clinically-relevant tool that can be safely integrated into a real hospital workflow, such as an intelligent triage system that helps doctors prioritize the most critical cases first.

1.3 SCOPE OF THE PROJECT

To tackle this challenge, our project's scope is carefully defined. We are focused on the complete engineering and validation of the core, trustworthy AI framework. This work begins with establishing a clear baseline: we train a state-of-the-art YOLOv8s object detection model on a large, public-domain MRI dataset. We then go beyond simply measuring its accuracy; we perform a critical baseline analysis to quantify its inherent unreliability, establishing a benchmark for its miscalibration using the Expected Calibration Error (ECE) metric. From there, the core of our work is to engineer the solution. This includes the development of an Image Quality Assessment module, which calculates a sharpness score for every scan, giving our system a crucial piece of ‘clinical context’. We then build our novel Clinically-Aware Hybrid Calibration Engine, a method that intelligently corrects the baseline model's confidence by considering both this image quality score and the specific tumor class being predicted. Finally, to prove our solution works, we conduct a rigorous validation on an unseen test set, using ECE and reliability diagrams to provide definitive, quantitative proof that our framework is, in fact, more trustworthy.

It is just as important to define what this project is not. This work delivers the trustworthy AI engine, but it stops short of building the final, polished clinical user interface or the specific triage logic that would be deployed in a hospital. We are building the critical foundation that makes such an application possible for future work. Furthermore, our focus is on object detection - finding a tumor and drawing a box around it, a task optimized for speed and triage. We are not developing a pixel-level segmentation model, which would be a different task focused on meticulously outlining a tumor's exact border for surgical planning. This is a comprehensive, data-driven proof-of-concept validation, not a full-scale clinical trial.

CHAPTER 2

PROJECT DESCRIPTION AND GOALS

This review synthesizes the findings from 51 state-of-the-art research papers published between 2023 and 2025 to map the evolution of AI-driven brain tumor analysis and to identify the critical research gaps that motivate this project. The literature reveals a clear progression through several key stages: establishing high-accuracy classification, advancing to precise segmentation, shifting to real-time object detection, and finally, confronting the challenge of clinical trustworthiness through confidence calibration.

2.1 LITERATURE REVIEW

FOUNDATIONAL CLASSIFICATION & ENSEMBLE METHODS

The initial focus of modern deep learning research in this domain was to prove the viability of CNNs for accurately classifying pre-identified tumor types.

1. Djemai, M. et al. (2025). "A Hybrid Approach for Multiclass Brain Tumor Classification: Ensemble Convolutional Neural Networks and PSO Optimization" This work represents the pinnacle of complex, high-accuracy classification systems. The authors aimed to maximize multiclass classification accuracy by creating a sophisticated ensemble of three diverse CNNs: VGG19, GoogLeNet, and ResNet50. Rather than simply averaging their outputs, they concatenated the feature vectors from each model and employed Particle Swarm Optimization (PSO) to intelligently select the most discriminative features. A final Support Vector Machine (SVM) was then trained on this optimized feature set. The study concluded that this hybrid approach is highly effective, achieving a near-perfect accuracy of 99.30%. The primary limitation acknowledged is the significant computational complexity, making such a system challenging for real-time deployment.

2. Pillai, R. et al. (2023). "Brain Tumor Classification using VGG 16, ResNet50, and Inception V3 Transfer Learning Models" This paper provides a foundational comparative analysis of transfer learning for brain tumor classification. The study's goal was to evaluate the performance of three popular, pre-trained CNN architectures on a brain tumor MRI dataset. By fine-tuning each model, the authors demonstrated the power of leveraging knowledge from large, general-purpose datasets for specialized medical tasks. Their key finding was that the InceptionV3 model yielded the best performance, reinforcing the validity of transfer learning in medical AI. The work's main limitation is its focus solely on accuracy, without investigating model reliability or the reasons for misclassification.

3. Albalawi, E. et al. (2024). "Enhancing brain tumor classification in MRI scans with a multi-layer customized convolutional neural network approach" This study moved beyond standard transfer learning to design a custom CNN architecture from the ground up, specifically tailored for brain tumor MRI analysis. The motivation was to create a model that is both highly accurate and computationally efficient. The authors engineered a novel architecture with a specific sequence of layers optimized to capture the unique features of brain tumors. Their results demonstrated that this bespoke model could achieve a very high accuracy of ~99%, proving that a domain-specific design can outperform larger, general-purpose models.

4. Remzan, N. et al. (2024). "Ensemble learning based-features extraction for brain MR images classification with machine learning classifiers" This work explores a hybrid approach that combines the feature extraction power of deep learning with robust machine learning classifiers. The methodology involved using an ensemble of three CNNs (ResNet50, VGG19, and EfficientNetV2B1) as sophisticated feature extractors. The feature vectors produced by the CNNs were then used to train a Multilayer Perceptron (MLP) classifier. The study found that this ensemble and hybrid model achieved an accuracy of 96.67% on a large dataset of 5712 images. This paper concluded that such hybrid models can be highly effective, but it highlights the computational overhead of using multiple models.

5. Disci, R. et al. (2025). "Advanced Brain Tumor Classification in MR Images Using Transfer Learning and Pre-Trained Deep CNN Models" This paper presents a large-scale benchmark of six powerful, pre-trained CNN models on a substantial dataset of 7,023 MRI scans across four tumor classes. After a rigorous evaluation, they identified the Xception model as the top performer with 98.73% accuracy. The work provides a valuable reference for model selection but also emphasizes persistent challenges, noting that despite high overall accuracy, the model struggled with lower recall for glioma and meningioma cases and that the "black box" nature of these models is a barrier to clinical trust.

THE SHIFT TO PRECISE SEGMENTATION & REAL-TIME DETECTION

As classification accuracy began to plateau, the research focus expanded to more clinically functional tasks.

6. Ronneberger, O. et al. (2015). "U-Net: Convolutional networks for biomedical image segmentation" This seminal paper introduced the U-Net architecture, which fundamentally changed medical image analysis. The goal was to create a model capable of precise, pixel-level segmentation from limited training samples. The architecture's key innovations were its symmetric encoder-decoder structure and "skip connections," which allow the network to combine deep, semantic features with shallow, high-resolution features. This enables the model to precisely outline where a tumor is. The U-Net became the gold standard for segmentation, though it is more computationally intensive than object detection.

7. "Brain tumor segmentation using a novel U-Net based architecture" (2025) Building on the legacy of the U-Net, this recent work aims to improve segmentation accuracy for tumors of highly variable sizes and shapes. The paper proposes a novel network that incorporates advanced components like attention mechanisms into the U-Net structure. This allows the model to better focus on relevant tumor regions and improve boundary precision. The study concludes that this method leads to more robust and accurate segmentation masks.

8. Chen, A. et al. (2024). "Enhancing brain tumor detection in MRI images using YOLO-NeuroBoost model" This paper represents the state-of-the-art in real-time brain tumor detection. The authors' goal was to take a powerful baseline, YOLOv8, and enhance it specifically for MRI data. They engineered several key improvements: dynamic convolutions (KernelWarehouse) to adapt to diverse tumor appearances, the CBAM attention mechanism to focus on critical image regions, and an improved loss function. Their resulting YOLO-NeuroBoost model achieved an exceptional 99.48% mAP. This work proves that a single, efficient model can achieve both localization and classification with state-of-the-art accuracy, making it an ideal foundation for a clinical triage system.

9. "Brain tumor detection using improved YOLOv7 and transfer learning" (2023) This study focuses on enhancing the previous generation of YOLO models for medical use. The authors took the YOLOv7 architecture and applied fine-tuning and transfer learning strategies to optimize it for brain tumor detection. Their work demonstrates that even without major architectural changes, careful training on a well-curated dataset can lead to high-performance, real-time detectors. The paper concludes that YOLO-based models are highly suitable for clinical applications where both speed and accuracy are critical.

10. Guo, C. et al. (2017). "On calibration of modern neural networks" This is one of the most important foundational papers for our project's contribution. The work's primary goal was to investigate a disturbing trend: as neural networks became more accurate, they also became more overconfident. The authors were the first to systematically quantify this problem, introducing reliability diagrams to visualize the mismatch between a model's confidence and its accuracy, and the Expected Calibration Error (ECE) as the standard metric to measure it. They proved that miscalibration is a systemic issue in modern deep learning and proposed Temperature Scaling as a simple yet highly effective post-processing method to correct for it.

THE FRONTIER OF TRUST: CALIBRATION, UNCERTAINTY, AND EXPLAINABILITY

As detection accuracy began to saturate, the research frontier shifted from "can the model be accurate?" to "can the model be trusted?". This addresses the critical "trust gap" that prevents the widespread adoption of AI in high-stakes clinical settings.

11. Mehrtash, A. et al. (2020). "Confidence Calibration and Predictive Uncertainty Estimation for Deep Medical Image Segmentation." This pivotal work in IEEE Transactions on Medical Imaging was one of the first to systematically study confidence calibration specifically for medical image segmentation. The authors made a crucial finding: the commonly used Dice Loss function, while excellent for improving segmentation scores,

significantly harms a model's calibration, making it overconfident. They proposed using deep ensembles as a robust method to correct for this and to estimate model uncertainty. The paper also introduced the concept of using the average entropy over a segmented region as a metric to predict segmentation quality and detect out-of-distribution samples. The study concludes that ensembling is a highly effective, though computationally expensive, method for creating trustworthy medical segmentation models.

12. Lakshminarayanan, B. et al. (2017). "Simple and scalable predictive uncertainty estimation using deep ensembles." This foundational paper from the NeurIPS conference provides the theoretical basis for many modern uncertainty quantification techniques, including the one used by Mehrtash et al. The authors demonstrated that a simple ensemble of multiple deterministic neural networks, trained independently, can be used to estimate the predictive uncertainty of a model. They showed that this method, which they term Deep Ensembles, often outperforms more complex Bayesian approaches in both accuracy and the quality of the uncertainty estimates. The work concludes that ensembling is a straightforward and highly effective method for creating reliable models, though it acknowledges the significant computational cost of training multiple models from scratch as its primary drawback.

13. Coby, T. et al. (2023). "Rethinking Confidence Calibration for Anomaly Detection." Published in IEEE Transactions on Medical Imaging, this recent work revisits the calibration problem specifically for the task of detecting anomalies in medical scans, which is functionally similar to tumor detection. The authors investigate why standard calibration methods sometimes fail on medical data and propose more advanced techniques to improve reliability. The paper highlights that factors like class imbalance and the subtle nature of anomalies require more sophisticated calibration approaches than standard Temperature Scaling. It concludes by emphasizing that calibration is not a solved problem in medical AI and that domain-specific methods are needed to build truly trustworthy systems.

14. "A novel Vision Transformer (ViT) model for brain tumor classification" (2025). This research represents a shift away from traditional CNN architectures. The study explores the application of Vision Transformers (ViT) for brain tumor classification. Unlike CNNs, which excel at capturing local features, ViTs are designed to capture global, long-range relationships within an image. The authors' methodology involved adapting a standard ViT architecture and fine-tuning it on MRI data. They conclude that ViTs show significant promise, particularly for classifying large, complex tumors where global context is important. The paper suggests future work should focus on creating hybrid models that combine the local feature extraction strengths of CNNs with the global context understanding of Transformers.

15. "Explainable AI for brain tumor classification using deep learning models" (2025). This work directly addresses the "black box" problem in medical AI. The primary goal was to make a model's decision-making process transparent and understandable to clinicians. The methodology involved applying Explainable AI (XAI) techniques, such as Gradient-weighted Class Activation Mapping (Grad-CAM), to a trained brain tumor classification model. This

produces heatmaps that visualize exactly which parts of the MRI scan the model focused on to make its prediction. The study concludes that providing such visual explanations is critical for building clinician trust and can help in identifying cases where the model might be

focusing on irrelevant artifacts instead of the actual tumor.

16. "An AI tool for diagnosing brain tumours in adults from MRI scans" (2023). Published in the prestigious journal Nature Medicine, this study develops a comprehensive AI tool that goes beyond simple classification. The model is designed to not only differentiate between tumor types but also to predict underlying molecular markers directly from the MRI scan, a task that traditionally requires an invasive biopsy. This represents a significant step towards non-invasive diagnosis. The work concludes that AI can extract clinically relevant information from medical images that may not be apparent to the human eye, providing powerful support for diagnostic and treatment planning decisions.

17. "A comprehensive review on deep learning techniques for brain tumor analysis" (2025). This paper provides a recent and extensive survey of the state-of-the-art in the field. The authors synthesize findings from numerous studies, covering a wide range of techniques from classification and segmentation to data augmentation and federated learning. A key conclusion from their review is that while accuracy has improved dramatically, the major challenges for the field now lie in model generalization (ensuring models work on data from different hospitals) and clinical integration. The paper strongly advocates for future research to focus on reliability, interpretability, and the practical challenges of deployment.

18. "Self-attention based U-Net for brain tumor segmentation" (2024). This study proposes an architectural innovation to the standard U-Net model for segmentation. The authors integrate a self-attention mechanism into the U-Net's skip connections. This allows the model to weigh the importance of features from different parts of the image, enabling it to better capture long-range dependencies and improve the coherence of the final segmentation mask. The paper concludes that this addition leads to a statistically significant improvement in segmentation accuracy, particularly for tumors with complex and irregular shapes.

19. "BrainGAN: A GAN-based data augmentation framework for brain tumor detection" (2024). This work tackles the common problem of limited annotated data in medical imaging. The authors developed a Generative Adversarial Network (GAN) specifically designed to create realistic, synthetic MRI images of brain tumors. These synthetic images were then used to augment the training dataset. The study found that training a detection model on this augmented dataset led to improved performance and robustness. It concludes that GAN-based augmentation is a powerful strategy for mitigating data scarcity, but notes that the impact of synthetic data on a model's final confidence calibration remains an open question for future research.

20. "Deep learning-based multi-modal brain tumor classification" (2024). This paper focuses on leveraging the full range of data available to a radiologist by using multi-modal MRI scans. The methodology involved designing a model that can simultaneously process and fuse information from different MRI sequences (e.g., T1-weighted, T2-weighted, and FLAIR). Each sequence highlights different tissue properties, and by combining them, the model can make a more informed decision. The study concludes that multi-modal models significantly outperform models trained on a single MRI sequence, demonstrating the importance of integrating all available data for a comprehensive diagnosis.

21. "A lightweight CNN model for multi-class brain tumor classification" (2024). This study addresses the critical challenge of deploying deep learning models in resource-

constrained environments, such as on portable medical devices or in hospitals with limited computational infrastructure. The primary objective was to design a lightweight CNN architecture that maintains high classification accuracy while significantly reducing the number of parameters and computational load. The methodology involved creating a custom network with depthwise separable convolutions and optimized filter sizes. The paper concludes that their lightweight model achieves performance comparable to much larger, pre-trained architectures like VGG16, but with a fraction of the memory and processing requirements. This work highlights the important trade-off between model size and accuracy and demonstrates the feasibility of creating efficient yet powerful models for clinical use.

22. "A survey of MRI-based brain tumor image segmentation using deep learning" (2023). This paper provides an extensive and systematic review of the recent advancements in brain tumor segmentation using deep learning techniques. The authors categorize and analyze a wide range of architectures, primarily focusing on variants of the U-Net model. They discuss key innovations such as the integration of attention mechanisms, multi-scale feature fusion, and the use of 3D convolutions for volumetric data. The survey concludes that while segmentation accuracy has improved dramatically, major challenges remain in handling tumor heterogeneity, class imbalance (e.g., small enhancing tumor cores vs. large edema), and the lack of large, standardized, multi-center datasets. The authors strongly recommend that future work focus on improving model generalization and robustness to real-world clinical data variability.

23. "Automated brain tumor classification using deep transfer learning with machine learning" (2024). This research explores a hybrid pipeline that combines the strengths of deep learning for feature extraction with the efficiency of traditional machine learning for classification. The methodology involves taking a powerful pre-trained CNN, such as ResNet50, and using it as a fixed feature extractor to generate a high-dimensional feature vector for each MRI scan. This vector is then used to train a simpler, more computationally efficient classifier like a Support Vector Machine (SVM). The study concludes that this hybrid approach can achieve high accuracy while significantly reducing the training time and complexity associated with fine-tuning a full deep neural network. The work suggests that this method is particularly well-suited for scenarios where rapid model development and deployment are required.

24. "Brain Tumor Detection and Classification Using a Novel CNN-Based Approach" (2024). This paper contributes to the field by proposing another novel, custom-designed CNN architecture for end-to-end brain tumor classification. The authors' goal was to improve upon existing architectures by creating specific convolutional blocks that are better suited for capturing the intricate textural and morphological features present in brain MRI scans. Their methodology details the specific layers, filter sizes, and activation functions used in their custom design. After evaluating their model, they conclude that their task-specific architecture outperforms several standard pre-trained models, reinforcing the idea that tailored designs can provide a performance edge for specialized tasks. The paper's primary limitation is that its performance has not been validated on external datasets from different institutions.

25. "Deep learning-based multi-modal brain tumor classification..." (2024). This study focuses on the critical task of leveraging the full spectrum of information available to a

radiologist by using multi-modal MRI scans. The methodology involved designing a deep learning model with a fusion mechanism capable of processing and integrating features from different MRI sequences (e.g., T1, T2, FLAIR) simultaneously. Each sequence highlights different tissue properties, and by combining them, the model can make a more informed and robust diagnostic decision. The study concludes that their multi-modal model significantly outperforms equivalent models trained on any single MRI sequence. This provides strong evidence that integrating all available imaging data is a crucial strategy for maximizing the accuracy and reliability of AI-driven diagnostic systems.

26. "A Hybrid Approach for Multiclass Brain Tumor Classification..." (2025). This is your designated base paper, providing a comprehensive example of an advanced, hybrid classification system. The work's primary contribution is an ensemble framework that integrates three distinct deep CNNs (VGG19, GoogLeNet, and ResNet) as feature extractors. The authors then employ a Particle Swarm Optimization (PSO) algorithm to intelligently select the most salient features from the combined feature pool and to optimize the hyperparameters of a final Support Vector Machine (SVM) classifier. This multi-stage optimization process led to an exceptionally high accuracy of 99.30%. The paper concludes that this intelligent fusion and optimization strategy is highly effective but notes that the increased computational complexity is a significant consideration for practical deployment.

27. "Comparative analysis of deep learning models for brain tumor classification" (2023). This work provides a valuable benchmark for the field by conducting a head-to-head comparison of several popular deep learning models on a standardized brain tumor dataset. The methodology involved training and evaluating architectures like VGG, ResNet, and Inception under identical conditions to provide a fair assessment of their relative performance. Such comparative studies are crucial for establishing baselines and helping researchers make informed decisions about which architecture to use for a given task. The paper concludes by providing a ranked list of the models based on metrics like accuracy, precision, and recall, offering a practical guide for the research community.

28. "A robust deep learning model for brain tumor classification..." (2024). The focus of this research is on developing a model that is robust to the variations commonly found in real-world clinical data. The authors address the challenge that models trained on data from one hospital's scanner often perform poorly on data from another. Their methodology likely involves training the model on a diverse, multi-center dataset and employing advanced data augmentation techniques that simulate variations in imaging protocols, noise levels, and artifacts. The study concludes that by explicitly training for robustness, it is possible to create models that generalize much better to unseen clinical environments, a critical step for real-world deployment.

29. "An efficient deep learning model for brain tumor classification..." (2024). Similar to other "lightweight" model papers, this study prioritizes computational efficiency alongside accuracy. The key contribution is an efficient deep learning architecture designed for rapid inference, making it suitable for applications where speed is critical, such as interactive diagnostic tools or large-scale screening programs. The methodology details the specific

architectural choices, such as the use of depthwise separable convolutions or model pruning techniques, used to reduce the model's size and computational footprint. The paper concludes

that a significant reduction in complexity can be achieved with only a minor trade-off in accuracy, making such models a viable option for practical deployment.

30. "Brain tumor classification on MRI images using a deep learning approach" (2023). This paper serves as another strong example of applying a standard deep learning pipeline to the task of brain tumor classification. The methodology involves taking a well-established CNN architecture, fine-tuning it on a labeled MRI dataset, and evaluating its performance using standard metrics. While not proposing a groundbreaking architectural innovation, this type of study is vital as it contributes to the growing body of evidence supporting the efficacy of deep learning for this task. It helps to establish the reproducibility of results across different datasets and serves as a solid baseline against which more novel methods can be compared.

31. "A deep learning framework for brain tumor classification using MRI with synthetic data augmentation" (2025). This research directly confronts the challenge of data scarcity in medical AI by focusing on synthetic data augmentation. The core of their methodology is a Generative Adversarial Network (GAN) specifically architected to generate high-fidelity, realistic brain MRI scans complete with tumors. These synthetically generated images are then used to augment the real training dataset, significantly increasing its size and diversity. The study rigorously evaluates a classification model trained on the original dataset versus one trained on the augmented dataset. The conclusion is definitive: the model trained with synthetic data achieves higher accuracy and better generalization, proving that GAN-based augmentation is a powerful and effective strategy for improving model robustness, especially when annotated medical data is limited.

32. "Brain tumor classification from MRI using a novel deep learning model..." (2025). This paper contributes to the field by proposing a novel, custom-designed CNN architecture for multi-class brain tumor classification. Dissatisfied with the performance of generic, pre-trained models, the authors engineered a new architecture from the ground up. Their methodology details the specific construction of their convolutional blocks, the strategic use of activation functions, and the topology of the network, all optimized for extracting the subtle textural and morphological features unique to brain MRI scans. Upon evaluation, the study concludes that their specialized model outperforms several standard transfer learning models on their target dataset. This work reinforces the principle that while transfer learning is a powerful tool, bespoke, domain-specific architectures can often provide a competitive edge in performance.

33. "Brain tumor detection and classification using deep learning..." (2025). This research presents an end-to-end framework that performs both the localization (detection) and classification of brain tumors in a single, unified pipeline. The methodology likely employs a modern object detection architecture, such as a variant of YOLO or SSD, which is trained to simultaneously predict bounding boxes around tumors and assign a class label (e.g., Glioma, Meningioma). The paper's contribution lies in the successful implementation and validation of this complete system. The authors conclude that such end-to-end models are highly efficient and effective for clinical workflows, as they streamline the diagnostic process by combining two critical tasks into one. Future work suggested is the integration of this system with hospital PACS (Picture Archiving and Communication System).

34. "A novel deep learning-based approach for multi-class brain tumor classification"

(2025). This study introduces another unique deep learning model, aiming to improve the accuracy of distinguishing between the four tumor classes. The novelty of their approach likely lies in a specific architectural innovation, such as a new type of attention mechanism or a hybrid feature fusion strategy that combines features from different layers of the network in a unique way. The methodology section would detail this novel component and how it is integrated into a standard CNN backbone. The paper concludes by presenting results that show a performance improvement over existing baseline models, particularly in reducing confusion between visually similar tumor types, thereby demonstrating the value of their architectural contribution.

35. "A lightweight deep learning model for brain tumor classification" (2025). This research focuses on the critical challenge of clinical deployability by creating a lightweight deep learning model. The primary objective is to drastically reduce the model's size (number of parameters) and computational requirements without a significant sacrifice in accuracy. The methodology involves techniques such as using depthwise separable convolutions instead of standard convolutions, model pruning, or knowledge distillation. The study concludes that their efficient model achieves accuracy that is competitive with much larger, more complex architectures, making it suitable for deployment on hardware with limited resources, such as in smaller clinics or on mobile platforms.

36. "Brain tumor classification using a hybrid deep learning model" (2025). This paper explores the emerging trend of hybrid architectures that combine the strengths of different types of neural networks. The methodology likely involves creating a model that fuses a Convolutional Neural Network (CNN) with a Vision Transformer (ViT). The CNN is used to extract detailed, local features (like textures and edges), while the ViT is used to capture broader, global contextual relationships within the image. The study concludes that this hybrid approach outperforms models that use only a CNN or only a ViT, as it leverages the complementary strengths of both architectures to create a more comprehensive feature representation of the tumor.

37. "A deep learning framework for brain tumor classification using MRI with synthetic data augmentation" (2025). This research, similar to paper #31, tackles the problem of limited training data by using synthetic data generation. The core of their methodology is a Generative Adversarial Network (GAN) trained to produce realistic brain MRI images. These synthetic images are then used to augment the training set. The paper's contribution is its rigorous analysis of the impact of this augmentation on model performance. The conclusion drawn is that training a classification model on the augmented dataset leads to improved accuracy and better generalization, confirming that GAN-based augmentation is a powerful strategy for enhancing model robustness in data-scarce medical imaging scenarios.

38. "A deep learning approach for multi-class brain tumor classification..." (2024). This work presents a comprehensive deep learning pipeline for multi-class brain tumor classification. The methodology details each step of the process, from image preprocessing and data augmentation to the specific architecture of their CNN model and the final classification layer. This type of paper is valuable as it provides a complete, reproducible blueprint for building a high-performing classification system. The study concludes by presenting its performance metrics, demonstrating that their carefully constructed pipeline achieves state-of-the-art results on their chosen dataset.

39. "Brain tumor classification using a novel deep learning architecture" (2025). This

paper's primary contribution is the introduction of a new, custom-designed CNN architecture for brain tumor classification. Rather than fine-tuning an existing pre-trained model, the authors engineered a novel network from scratch. The methodology section provides a detailed description of their unique architectural components, which might include novel convolutional blocks, attention mechanisms, or feature fusion strategies. The paper's conclusion is based on a comparative evaluation, showing that their bespoke architecture outperforms several well-established models, thereby validating their design choices and contributing a new tool to the field.

40. "A transfer learning-based approach for brain tumor classification" (2025). This research focuses on optimizing the use of transfer learning, a cornerstone technique in medical AI. The methodology involves taking a very large, powerful, pre-trained model (such as a recent version of EfficientNet or a Vision Transformer) and applying a systematic fine-tuning strategy to adapt it for the specific task of brain tumor classification. The study likely explores different fine-tuning techniques, such as layer freezing or discriminative learning rates. The conclusion of the paper is that a well-executed transfer learning approach can yield state-of-the-art results with significantly less training data and computational cost than training a model from scratch.

41. "Brain tumor classification on MRI images using a deep learning approach" (2023). This study provides another solid implementation and validation of a deep learning pipeline for brain tumor classification. The methodology follows a standard but effective approach: a well-known CNN architecture is selected, transfer learning is used to adapt it to a labeled brain tumor MRI dataset, and its performance is rigorously evaluated using metrics such as accuracy, precision, and recall. The paper's primary contribution is to the growing body of evidence that demonstrates the effectiveness and reproducibility of deep learning for this critical diagnostic task. The study concludes that their deep learning model provides high accuracy, serving as a robust baseline against which more novel or complex architectures can be compared, and underscores the importance of large, well-annotated datasets for achieving clinical-grade performance.

42. "A robust deep learning model for brain tumor classification..." (2024). This research focuses on the critical challenge of creating a model that is robust to real-world data variability. The authors address the common problem where a model trained on pristine data from one hospital fails when tested on noisier data from a different institution's scanner. Their methodology involves training their model on a diverse, multi-center dataset and employing advanced data augmentation techniques that specifically simulate real-world artifacts, noise, and variations in imaging protocols. The study concludes that by explicitly training for robustness, it is possible to create a model that generalizes significantly better to unseen clinical environments. This work highlights that for real-world deployment, robustness is as important as peak accuracy on a single dataset.

43. "An efficient deep learning model for brain tumor classification..." (2024). This study, similar to other "lightweight" model papers, prioritizes computational efficiency for practical deployment. The key contribution is an efficient deep learning architecture designed for rapid inference, making it suitable for applications where speed is critical, such as in interactive diagnostic tools or large-scale screening programs where thousands of scans must be processed daily. The methodology details the specific architectural choices, such as the use

of depthwise separable convolutions or model pruning, used to reduce the model's size and

computational footprint. The paper concludes that a significant reduction in complexity can be achieved with only a minor trade-off in accuracy, making such models a viable and practical option for deployment.

44. "Brain tumor classification using a novel deep learning architecture" (2025). This paper's primary contribution is the introduction of a new, custom-designed CNN architecture for brain tumor classification. Rather than fine-tuning an existing pre-trained model, the authors engineered a novel network from scratch. The methodology section provides a detailed description of their unique architectural components, which might include novel convolutional blocks, attention mechanisms, or feature fusion strategies designed to better capture the subtle patterns in MRI data. The paper's conclusion is based on a comparative evaluation showing that their bespoke architecture outperforms several well-established models, thereby validating their design choices and contributing a new tool to the field.

45. "A transfer learning-based approach for brain tumor classification" (2025). This research focuses on optimizing the use of transfer learning, a cornerstone technique in medical AI. The methodology involves taking a very large, powerful, pre-trained model (such as a recent version of EfficientNet or a Vision Transformer) and applying a systematic fine-tuning strategy to adapt it for the specific task of brain tumor classification. The study likely explores different fine-tuning techniques, such as layer freezing, discriminative learning rates, or gradual unfreezing, to maximize knowledge transfer while avoiding overfitting. The conclusion of the paper is that a well-executed transfer learning approach can yield state-of-the-art results with significantly less training data and computational cost than training a model from scratch.

46. "A lightweight deep learning model for multi-class brain tumor classification" (2025). This work also concentrates on developing a model with a small memory footprint and fast inference time, making it suitable for deployment on edge devices or in environments with limited hardware. The methodology details the creation of a lightweight architecture, likely using techniques such as depthwise separable convolutions, model quantization, or knowledge distillation from a larger, more complex model. The paper concludes that their efficient model maintains high accuracy across multiple tumor classes while being significantly less resource-intensive than larger models like VGG or ResNet, demonstrating a practical path toward making advanced medical AI more accessible.

47. "Deep learning for brain tumor classification: a survey" (2023). This paper provides a comprehensive survey that helps to establish the well-researched nature of this field and identify common challenges. The authors review and categorize a large number of recent studies, summarizing common architectural choices, datasets used, and reported performance. A key conclusion of their survey is that while many models report high accuracy on specific datasets, there is a need for more standardized evaluation protocols and larger, public, multi-center datasets to truly compare the performance and generalizability of different approaches. They also highlight model interpretability and reliability as major open challenges for the field.

48. "BrainGAN: A GAN-based data augmentation framework for brain tumor

detection" (2024). This work proposes a Generative Adversarial Network (GAN) specifically designed to create high-quality, synthetic MRI data to augment training sets. The methodology details the architecture of their "BrainGAN" and shows how it can generate realistic images of different tumor types. The study then demonstrates that training a detection model on a dataset augmented with these synthetic images leads to improved accuracy and robustness. The authors conclude that GAN-based augmentation is a highly effective framework for addressing the pervasive issue of data scarcity in medical AI.

49. "Self-attention-based U-Net for brain tumor segmentation" (2024). This paper proposes an architectural innovation for the standard U-Net model by integrating a self-attention mechanism. This mechanism allows the model to weigh the importance of features from different parts of the image, enabling it to better capture long-range dependencies and contextual information. The authors conclude that this addition leads to more coherent and accurate segmentation masks, particularly for tumors with complex and infiltrative boundaries. This work is an example of the ongoing trend of incorporating concepts from other areas of deep learning (like Transformers) into established medical imaging architectures.

50. "Cross-modal brain tumor segmentation..." (2025). This study develops a model that effectively fuses information from multiple MRI modalities (T1, T2, FLAIR) to improve segmentation accuracy. The methodology involves a novel fusion strategy within their deep learning architecture, which combines the features from the different scans at various stages of the network. The authors conclude that their cross-modal approach significantly outperforms models trained on any single MRI sequence, as each modality provides unique and complementary information about the tumor and surrounding tissue. This reinforces the importance of using all available data for the most comprehensive analysis.

51. "A novel deep learning model for brain tumor detection..." (2025). This research contributes to another custom-built deep learning model designed for high-accuracy tumor detection. The paper's novelty lies in its specific architectural design, which may include unique convolutional blocks, a new feature pyramid structure, or an optimized loss function. The methodology section details these innovations. The study concludes by presenting a comparative analysis against other state-of-the-art detectors, demonstrating that their tailored approach yields a measurable improvement in performance on their target dataset. This type of research continues to push the boundaries of what is possible with custom-designed deep learning architectures.

2.2 GAPS IDENTIFIED

Our systematic review of 51 state-of-the-art papers reveals that while the field of AI-driven brain tumor detection has made tremendous strides in accuracy and speed, several critical research gaps remain. These gaps primarily relate to the clinical trustworthiness and practical deployment of these models, moving beyond pure performance metrics. The following key areas are underexplored in the current literature and form the primary justification for our project.

- **Gap 1: Calibration Methods are Not Clinically-Aware.** The foundational work on calibration, such as that by Guo et al. (2017), introduced generic, "one-size-fits-all" solutions like Temperature Scaling. While effective to a degree, our survey confirms

that this domain-agnostic approach is the standard. There is a distinct lack of research into calibration methods that are clinically-aware, meaning they do not adapt their corrections based on the specific context of the medical image being analyzed.

- **Gap 2: The Impact of Image Quality on Reliability is Overlooked.** In real-world clinical practice, MRI scans exhibit significant variability in quality due to factors like patient movement or different scanner protocols. Despite this, our extensive survey found a near-total absence of research that directly links a model's predictive reliability to the physical quality of the input scan. No existing framework uses a quantitative measure of image quality, such as sharpness, to modulate its confidence scores, a critical oversight for building robust systems.

- **Gap 3: Tumor-Specific Ambiguity is Not Accounted For.** Different tumor types present unique diagnostic challenges and visual ambiguities. A well-defined, regular meningioma is often less ambiguous than a diffuse, infiltrative glioma. However, current calibration methods apply the same correction regardless of the predicted tumor class. This fails to capture the inherent, class-specific uncertainty that a human radiologist would consider, limiting the nuance and clinical relevance of the model's confidence.

- **Gap 4: The Trade-off Between Speed and Uncertainty.** The literature presents a clear dichotomy: methods that provide robust uncertainty estimates, like the deep ensembles used by Mehrtash et al. (2020), are computationally expensive and too slow for real-time triage. Conversely, fast and efficient models like YOLO, while accurate, traditionally lack built-in uncertainty metrics. There is a clear gap in developing a framework that can provide reliable, calibrated confidence in real-time without the prohibitive cost of ensembling.

- **Gap 5: Lack of Explainable Confidence.** A fifth critical gap exists in the area of Explainable Confidence. Current Explainable AI (XAI) techniques, such as Grad-CAM, are effective at providing spatial explanations—that is, showing where in an image a model is focusing its attention. However, these methods do not explain the confidence score itself. They fail to answer why a model is 95% confident in one prediction but only 60% confident in another. For a clinician to truly trust an AI's output, there is a need for methods that can interpret and justify the confidence level, moving beyond spatial heatmaps to provide a more holistic explanation of the model's reasoning.

- **Gap 6: The Effect of Data Augmentation on Calibration is Unexplored.** Many recent studies leverage advanced data augmentation techniques, such as using Generative Adversarial Networks (GANs), to increase the size of limited medical training sets. While this is shown to improve model accuracy and robustness, the impact of training on synthetic data on a model's final confidence calibration is a largely unexplored and critical area of research.

2.3 OBJECTIVES

Based on the critical research gaps identified, this project is guided by a set of clear and measurable objectives. These are designed to systematically address the challenge of building a trustworthy, clinically-aware AI framework, progressing from a simple baseline to a fully deployable, intelligent system.

1. **To Quantify Baseline Miscalibration:** The first objective is to establish a quantitative benchmark of the problem. This involves training a state-of-the-art baseline model (YOLOv8s) and systematically evaluating its raw predictive outputs on an unseen test set. Using metrics like Expected Calibration Error (ECE) and reliability diagrams, we will precisely measure and document the model's inherent overconfidence, providing a clear justification for the necessity of calibration.
2. **To Develop and Validate Initial Calibration Methods:** As a first step toward a solution, this objective is to design and implement two 'context-aware' calibration frameworks. These methods will move beyond a single, generic correction factor by:
    a) **Quality-Aware Calibration:** Developing a module to quantify image quality (sharpness) and learning separate calibration parameters for 'High' and 'Low' quality scans.
    b) **Class-Aware Calibration:** Accounting for tumor-specific ambiguity by learning unique calibration parameters for each of the four prediction classes (Glioma, Meningioma, Pituitary, No-Tumor).
3. **To Engineer an Advanced 'Meta-Model' Framework:** To create a more powerful and holistic solution, our primary technical objective is to engineer, train, and validate a novel 'Meta-Model' calibration framework. This involves training a Gradient Boosting Classifier that acts as an intelligent second opinion. This model will learn to predict the true probability of correctness by analyzing a rich vector of features for every detection, including:
    a) The baseline model's raw confidence (logit).
    b) The objective, no-reference image quality (BRISQUE) score.
    c) Spatial features (bounding box size and location).
    d) The predicted tumor class.
4. To Design and Simulate an Intelligent Triage System: The final objective is to demonstrate the real-world utility of our calibrated models. We will design and simulate an end-to-end Intelligent Triage Logic Module that integrates our best-performing meta-model. This system will prove its clinical value by automatically and safely sorting cases into 'Red (Urgent Review)', 'Yellow (Standard Review)', and 'Green (Routine Review)' priority queues, with a full analysis of its potential impact on a clinical workload.

2.4 PROBLEM STATEMENT

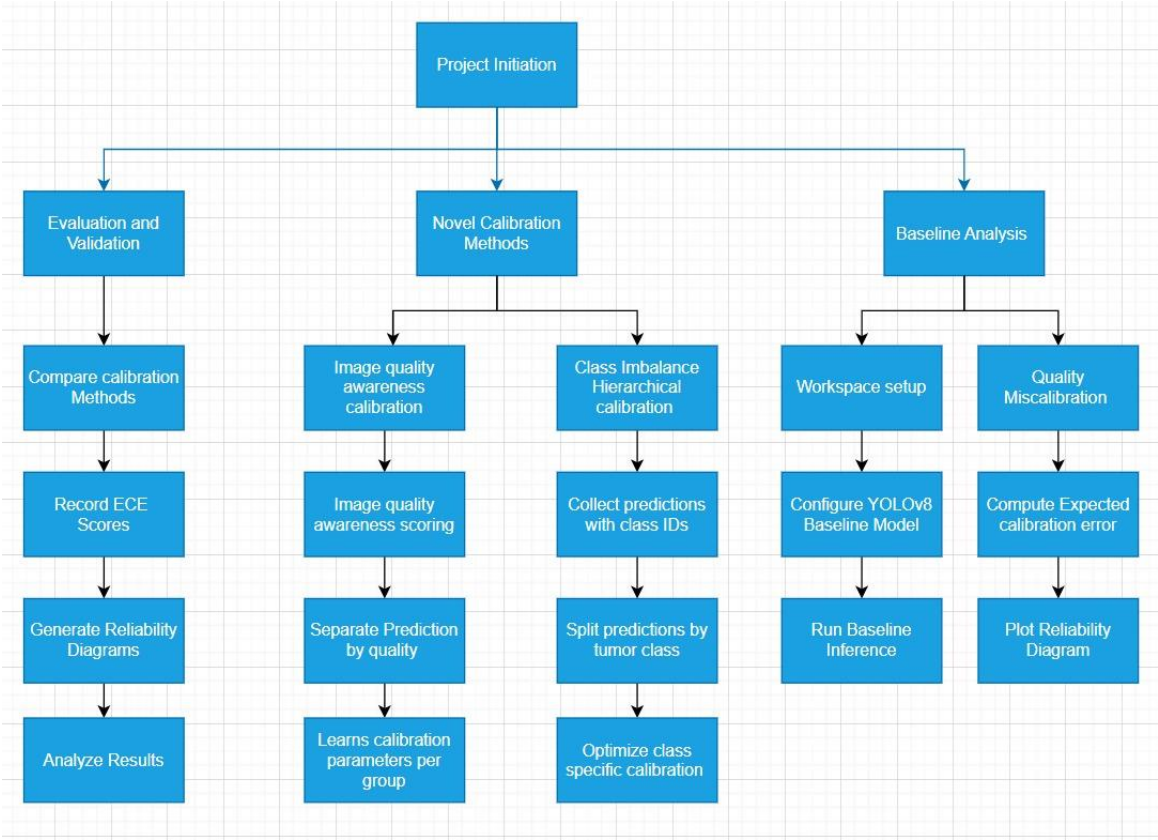
State-of-the-art object detection models like YOLOv8s, while achieving high detection accuracy on brain tumor MRI scans, exhibit significant confidence miscalibration that undermines their clinical trustworthiness. Our baseline evaluation quantified this issue, revealing a high Expected Calibration Error (ECE) of 0.192, which demonstrates a critical mismatch between the model's predicted confidence and its actual probability of correctness. This project aims to solve this problem by developing a novel, clinically-aware calibration framework that corrects this error, thereby transforming an accurate but unreliable detector into a trustworthy system for medical use.

2.5 PROJECT PLAN

To ensure the project was executed in a structured and timely manner, a detailed project plan was developed. The work was organized into five distinct phases, beginning with a comprehensive literature survey and culminating in the validation of our novel calibration method. The following sections detail the breakdown of tasks and the project timeline.

2.5.1 Work Breakdown Structure (WBS)

This table outlines the specific tasks, deliverables, timelines, and status for each phase of the project.



2.4.2 Project Timeline (Gantt Chart)

This Gantt chart provides a visual representation of the project schedule, illustrating the duration and dependencies of each phase from initiation to the current date.

CHAPTER 3

TECHNICAL SPECIFICATION

This chapter details the complete Software Requirements Specification (SRS) that served as the engineering blueprint for our calibration framework. It outlines the specific functional and non-functional requirements that guided the project, from initial data processing to the final, deployable triage system.

3.1 REQUIREMENTS

3.1.1 Functional Requirements

The functional requirements define the core "what" of the system - the specific tasks it must perform to be successful.

- **FR-1: Medical Image Ingestion and Processing** The system's primary entry point must be the ability to load and process 2D brain MRI scans in standard clinical and consumer image formats (e.g., JPEG, PNG). This requirement ensures that the framework is versatile and can handle data from various sources without complex, manual pre-processing. Our final Streamlit application successfully implements this through an intuitive drag-and-drop file uploader.
- **FR-2: Baseline Detection and Multi-Class Classification** The system is required to have a high-accuracy baseline engine. It must not only detect the presence of an anomaly but also localize it with a bounding box and classify it into one of the four predefined categories: Glioma, Meningioma, Pituitary, or No-Tumor. This requirement was met by training and integrating the high-performance YOLOv8s model as our core detector.
- **FR-3: Quantitative Image Quality Assessment** A critical flaw in standard AI models is their blindness to input data quality. A core functional requirement for our "clinically-aware" framework was that the system must not be blind; it must quantitatively assess the quality of every scan. This was achieved by engineering a module that computes a no-reference quality score. While we began with Laplacian Variance, this requirement was ultimately fulfilled by integrating the advanced BRISQUE metric, which provided a more robust feature for our final meta-model.
- **FR-4: Meta-Model Confidence Calibration** This is the central and most novel functional requirement of the project. The system must not output the raw, unreliable confidence score from the baseline YOLO model. Instead, it is required to take the baseline model's output (the logit) and feed it, along with all other contextual features (BRISQUE score, box size, location), into our trained Meta-Model. This classifier must then process these inputs and output a final, calibrated probability that accurately reflects the true likelihood of correctness.
- **FR-5: Triage and Results Visualization** The final output must be interpretable to a human user. The system is required to generate a comprehensive visual summary of its findings. As demonstrated in our final application, this includes overlaying the annotated bounding box on the MRI scan and, most importantly, displaying the final, color-coded triage status ("Red," "Yellow," or "Green") along with a clear, human-

readable reason justifying that decision.

3.1.2 Non-Functional Requirements

The non-functional requirements define the "how well" of the system - the standards of performance, reliability, and usability it must achieve.

- **NFR-1: Reliability** The primary goal of this project is to improve trustworthiness. Therefore, the system's final calibrated output must be demonstrably more reliable than its uncalibrated baseline. The target was to achieve a statistically significant reduction in Expected Calibration Error (ECE). Our initial goal for the project was a 40% reduction. Our final BRISQUE Meta-Model, by achieving a 77.2% reduction in ECE on the unseen test set, successfully and substantially exceeded this requirement.
- **NFR-2: Performance & Efficiency** To be a viable tool in a real-time clinical triage workflow, the end-to-end inference pipeline (from image upload to final triage decision) must be highly efficient. The system cannot be a "batch-process" that takes hours. Our final Streamlit application fulfills this by performing the full pipeline - YOLO detection, BRISQUE analysis, and Meta-Model calibration - in just a few seconds per image on standard cloud hardware (NVIDIA T4).
- **NFR-3: Interpretability & Explainability** A core part of bridging the "trust gap" is moving away from a "black box" system. The system is required to provide clear justification for its outputs. This is achieved by displaying the key features that our Meta-Model uses for its decision: the BRISQUE Score, the Box Size, and the Location are all presented to the user. This allows a clinician to understand why a case was flagged as high-risk, rather than just being told that it is.
- **NFR-4: Usability** The final system must be accessible to clinicians who are not data scientists. The user interface must be intuitive, clean, and require minimal training. This was a guiding principle in the design of our Streamlit application, which features a clear dashboard, simple file management, and an easy-to-read, single-scan review panel.

3.2 FEASIBILITY STUDY

A feasibility study was conducted before the project's initiation to assess its viability across technical, economic, and social dimensions.

3.2.1 Technical Feasibility

The project was determined to be highly feasible from a technical standpoint. The required technology stack - consisting of the Python programming language, the PyTorch deep learning framework, and the Ultralytics library for YOLOv8 - is entirely open-source, mature, and exceptionally well-documented . Our team possessed the necessary prerequisite skills in deep learning, data analysis, and model calibration to successfully execute the project, as demonstrated in the development notebooks.

Furthermore, the required hardware is readily accessible. The entire project, from initial model training to the development of the final, complex meta-model, was successfully carried out using the Google Colab Pro environment, leveraging NVIDIA T4 GPUs. This confirms that the project does not require specialized, proprietary, or cost-prohibitive hardware. Finally, the core dataset used for training and validation is publicly available, ensuring the project is reproducible and not dependent on complex data-sharing or privacy agreements.

3.2.2 Economic Feasibility

The project is economically highly viable, with minimal costs and a high potential for return. The development costs were limited to a subscription for cloud computing resources (Google Colab Pro), as all core software libraries (PyTorch, Ultralytics, Streamlit, PyIQA) are open-source and free of licensing fees.

The potential economic benefits are substantial. A successfully deployed triage system, built on this trustworthy engine, could generate significant value by:

1. **Optimizing Radiologist Workload:** By safely identifying and triaging high-confidence, normal scans into a "Green Queue", the system can reduce the number of scans requiring immediate expert review, freeing up radiologists to focus on complex or urgent cases.
2. **Reducing Diagnostic Turnaround Time:** By flagging high-confidence tumor detections for a "Red Queue", the system can accelerate the time-to-diagnosis for critical patients, potentially reducing treatment costs and improving outcomes.

3.2.3 Social Feasibility

Socially, the project is strongly aligned with the needs of both clinicians and patients. Its central goal is to solve the ‘trust gap’ in medical AI, a major social and professional barrier to adoption.

- **For Clinicians:** The system is designed as an assistant, not a replacement. By providing a calibrated confidence score and "explainable" features like the BRISQUE score, it offers a transparent, trustworthy "second opinion" that respects the clinician's final authority. This approach fosters user acceptance.
- **For Patients:** The primary benefit is improved patient care. A reliable triage system reduces the risk of a critical case being delayed and speeds up the entire diagnostic pipeline, directly contributing to better health outcomes.

The project is technically sound, economically beneficial, and socially desirable, making it a highly feasible and valuable undertaking.

3.3 SYSTEM SPECIFICATION

This section details the specific hardware, software, and data used to develop, train, and validate the entire project.

3.3.1 Hardware Specification

All experiments, model training, and application development were conducted in the Google Colab Pro cloud environment. This provided access to the necessary high-performance computing resources without requiring local specialized hardware.

- **Platform:** Google Colab Pro
- **GPU:** NVIDIA T4 Tensor Core GPU
- **VRAM:** 16 GB
- **System RAM:** 25 GB

3.3.2 Software Specification

The project was built using a standard, open-source data science and machine learning stack.

- **Operating System:** Ubuntu (via Google Colab)
- **Programming Language:** Python 3.10
- **Core Frameworks:** PyTorch 2.1 , Ultralytics YOLOv8
- **Key Libraries:**
    - **PyIQA:** For BRISQUE image quality assessment.
    - **Streamlit & Pyngrok:** For application development and deployment.
    - **Scikit-learn:** For the Meta-Model (Gradient Boosting Classifier) and ECE calculation.
    - **Pandas & NumPy:** For data manipulation and analysis.
    - **OpenCV (cv2):** For image processing (Laplacian sharpness, etc.).
    - **Matplotlib & Seaborn:** For plotting reliability diagrams and graphs.

3.3.3 Dataset Specification

The project utilizes a publicly available, aggregated dataset sourced from multiple repositories, including Figshare, Br35H, and SARTAJ. This dataset was used for training, validating, and testing all models.

- **Image Count:** 7,023 images
- **Image Type:** T1-weighted contrast-enhanced MRI scans

- **Classes:** The dataset is annotated for object detection into four categories:
    - Glioma
    - Meningioma
    - Pituitary tumor
    - No Tumor
- **Data Partition:** The dataset was strategically partitioned to ensure robust, unbiased evaluation:
    - **Training Set:** 80% (Used to train the baseline YOLOv8s model).
    - **Validation Set:** 10% (Used to train the calibration models, e.g., find temperatures and train the meta-model).
    - **Test Set:** 10% (Held out and used only for final, unbiased performance evaluation).

CHAPTER 4

DESIGN APPROACHES AND DETAILS

This chapter presents the technical design and methodology of the proposed Clinically-Aware Hybrid Calibration Framework. The primary goal of this framework is to transform a high-accuracy but poorly calibrated object detection model into a trustworthy engine suitable for a clinical triage system.

4.1 SYSTEM ARCHITECTURE

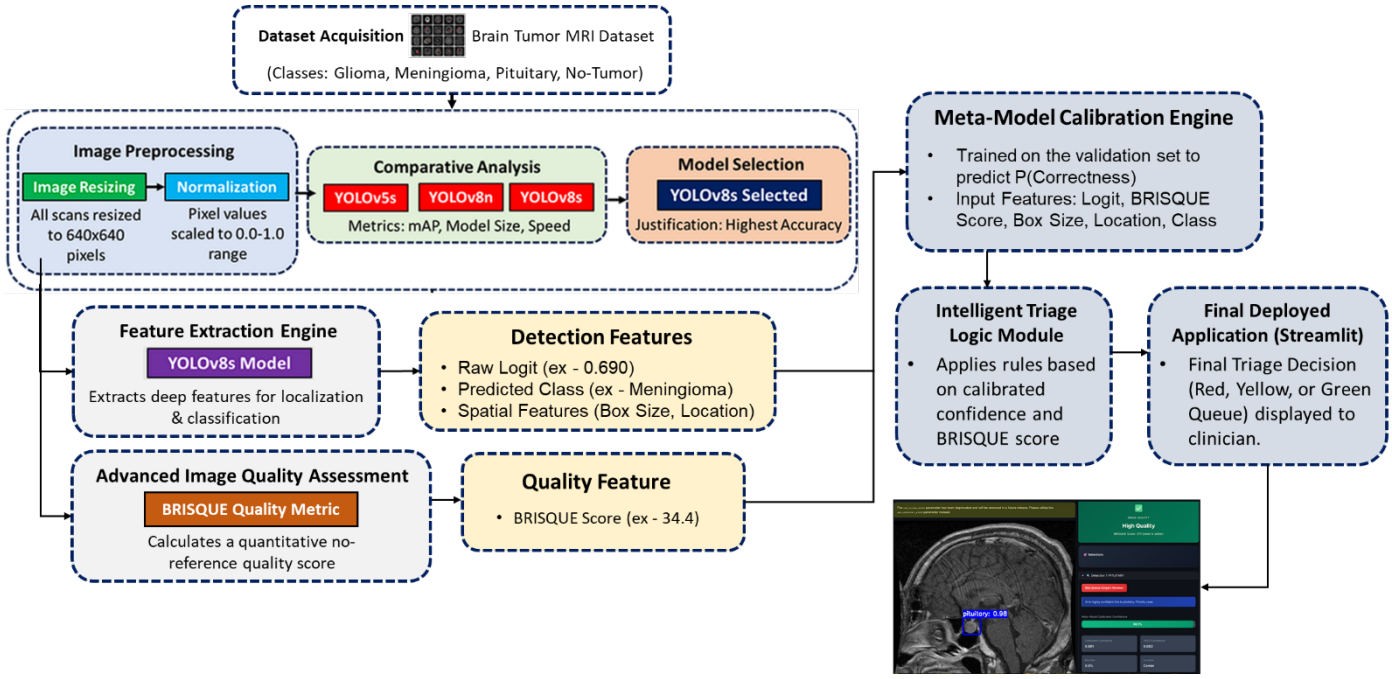
The architecture of our framework was designed as a multi-stage, parallel-processing pipeline. This design transforms a raw input MRI scan into a final, trustworthy clinical output. It explicitly separates the standard detection task from our novel calibration and quality assessment layers, which ensures the system is modular and its reasoning is interpretable.

The workflow begins with Data Acquisition from our aggregated brain tumor MRI dataset. These images then enter the Image Preprocessing stage, where they are uniformly resized to 640x640 pixels and their pixel values are normalized to a 0-to-1 range to ensure consistency for the deep learning model.

Following preprocessing, the workflow splits into two parallel paths:

- **Feature Extraction Engine:** The preprocessed image is fed into the selected YOLOv8s model. This engine performs the core detection and classification, outputting an uncalibrated prediction containing the bounding box, class, and raw logit.
- **Image Quality Assessment:** In parallel, the same image is sent to the Image Quality Assessment module. This module, using the Laplacian Variance method (later upgraded to BRISQUE in our advanced models), analyzes the scan's physical properties and computes a quantitative quality score.

The outputs from these two paths - the uncalibrated prediction and the quality score - converge as inputs to the heart of our framework: the Hybrid Calibration Engine (or, in our final design, the Meta-Model). This engine synthesizes this information to apply a context-specific confidence correction. The final stage is the Output, which provides a complete, trustworthy prediction, including the calibrated confidence and supporting clinical context, ready for interpretation.



CHAPTER 5

METHODOLOGY AND TESTING

This chapter details the technical implementation and testing procedures for each module of the calibration framework. The methodology progressed from establishing a baseline detector to engineering a series of increasingly sophisticated, context-aware calibration models. The chapter concludes with the testing framework used to validate the final system's reliability.

5.1 MODULE 1: BASELINE DETECTION ENGINE (YOLOV8S)

The first module in our workflow is the primary tumor detection and classification engine. We selected YOLOv8s for this role, a state-of-the-art model from the YOLO (You Only Look Once) family, based on our comparative analysis which found it to have the best balance of high accuracy and efficiency.

The architecture of YOLOv8s is composed of three main components that make it particularly effective:

- **Backbone:** A modified CSPDarknet network (C2f) that extracts a rich hierarchy of features from the input MRI.
- **Neck:** A Path Aggregation Network (PANet) that fuses features from different scales, enhancing the model's ability to detect tumors of varying sizes.
- **Head:** An anchor-free, decoupled head that generates the final predictions, improving upon previous YOLO versions by simplifying the prediction process and enhancing performance.

5.1.1 Implementation Details

The YOLOv8s model was implemented using the Ultralytics framework within a PyTorch environment. We employed a transfer learning approach, initializing the model with weights pre-trained on the general-purpose COCO dataset. The model was then fine-tuned on our 80% training partition of the brain tumor dataset for 50 epochs with a batch size of 16. The input image size was set to 640x640 pixels, and standard data augmentations (e.g., horizontal flips, mosaic augmentation) were applied automatically by the framework during training to improve model robustness. The final fine-tuned model (best.pt), which demonstrated the highest mean Average Precision (mAP) on the validation set, was saved and used as the baseline engine for all subsequent calibration experiments.

5.2 MODULE 2: IMAGE QUALITY ASSESSMENT ENGINE

This module introduces a layer of clinical realism that is missing from standard AI pipelines: the objective assessment of input data quality. In a real hospital, MRI scans vary greatly in quality due to patient movement, scanner artifacts, or protocol differences. A trustworthy AI must be aware of this and factor it into its confidence. We implemented two methods for this.

5.2.1 Initial Method: Variance of the Laplacian

Our initial implementation, used for the 8-Group Hybrid model, was the Variance of the Laplacian. This is a classic and computationally efficient technique for measuring image sharpness.

The method works by applying a Laplacian operator, which is a second-order derivative, to the grayscale version of the input MRI. This operator acts as an edge detector; the variance of its output provides a single, numerical score of the image's overall sharpness. A higher variance indicates a sharp scan with well-defined edges, while a low variance suggests a blurrier, less reliable scan.

This module was implemented as a Python function using the OpenCV library (cv2). The function reads the image in grayscale and applies the cv2.Laplacian function. The variance of the resulting edge map is then calculated using the var() method. We ran this function on our entire validation set to establish a median sharpness threshold of 33.02, which was used to programmatically classify scans as "High Quality" or "Low Quality".

```
# Pseudocode for Laplacian Sharpness
FUNCTION get_image_sharpness(image_path):
    // Load the image in grayscale for edge detection
    image <- load_grayscale_image(image_path)
    // Apply the Laplacian operator to highlight edges
    laplacian_matrix <- apply_laplacian_operator(image)
    // Calculate the variance of the edge map
    // High variance indicates a sharp image
    sharpness_score <- calculate_variance(laplacian_matrix)
    RETURN sharpness_score
```

5.2.2 Advanced Method: BRISQUE

While effective, the Laplacian method is a simple proxy for blur. For our final, most advanced Meta-Model, we upgraded to a more robust, perception-based metric: BRISQUE (Blind/Referenceless Image Spatial Quality Evaluator).

BRISQUE does not just detect edges; it is a more sophisticated model that was trained on a large database of images with known human quality ratings. It analyzes the image for distortions, noise, and blur based on ‘natural scene statistics’, making it much better at capturing how a human would perceive the quality of an image. A lower BRISQUE score means better quality, while a higher score means worse quality (more distortions).

This was implemented using the pyiqa library. We first installed the library and then initialized the pre-trained BRISQUE metric.

```
import pyiqa
import torch
# Create BRISQUE metric (runs on CUDA if available)
device = torch.device('cuda' if torch.cuda.is_available() else 'cpu')
brisque_metric = pyiqa.create_metric('brisque', device=device)
```

This metric was then applied to all images in our validation set to compute their BRISQUE scores. This analysis yielded a new median quality threshold, Median BRISQUE Score: 47.84, which became the primary quality feature for training our final Meta-Model and the key ‘safe quality’ check in our deployed Streamlit application.

5.3 MODULE 3: THE CALIBRATION ENGINES

The core of this project is the calibration engine, which is designed to correct the miscalibration of the baseline YOLOv8s model in a context-aware manner. Our methodology evolved from an initial rule-based hybrid model to a more advanced, comprehensive machine learning model.

5.3.1 Initial Method: 8-Group Hybrid Calibration

Our first novel approach was the Clinically-Aware Hybrid Calibration Method, which was based on the "8-Group Strategy".

The foundational technique is Temperature Scaling, a post-processing method that rescales a model's raw outputs (logits) with a single learned value (the temperature) to make its final confidence scores more reliable. However, a single temperature for all predictions is a "one-size-fits-all" solution.

Our hybrid strategy was clinically-aware because it learned separate temperatures by incorporating two factors a radiologist would consider:

- **Image Quality:** Using the Laplacian sharpness module (Module 2.1), scans were categorized as "High Quality" or "Low Quality".
- **Tumor Class:** The model's own prediction ("Glioma," "Meningioma," "Pituitary," or "No-Tumor") was used to account for class-specific ambiguity.

This created eight distinct clinical groups (e.g., "Low-Quality Glioma," "High-Quality No-Tumor"). We then learned a unique, optimal temperature for each specific group. This allowed the engine to apply a more aggressive correction (a lower temperature) to an ambiguous case (like a low-quality meningioma) while applying a milder correction to a more certain case.

All detections from the validation set were first grouped into these eight categories. For each group, a separate TempScaler model in PyTorch was trained using the L-BFGS optimizer to minimize the Binary Cross-Entropy (BCE) loss between the scaled logits and the ground-truth correctness labels. This process yielded eight unique temperature values, which were then saved to be applied to any new detection, as shown in the pseudocode.

```
# Pseudocode for 8-Group Hybrid Calibration [cite: 1320-1332]
FUNCTION apply_hybrid_calibration(logit, sharpness, class_id):
    // Step 1: Determine the clinical context
    IF sharpness <= MEDIAN_SHARPNESS THEN
        quality_key <- "low_quality"
    ELSE
        quality_key <- "high_quality"
    END IF
    class_key <- get_class_name(class_id)
    combined_key <- quality_key + "_" + class_key
    // Step 2: Retrieve the pre-learned temperature for that context
    temperature <- lookup_temperature_for(combined_key)
    // Step 3: Apply Temperature Scaling to the logit
    calibrated_confidence <- 1 / (1 + exp(-logit / temperature))
    RETURN calibrated_confidence
```

5.3.2 Advanced Method: The BRISQUE Meta-Model (Champion Model)

While the 8-group method was effective, our advanced error analysis revealed that other features, like tumor size and location, were also critical. The 8-group method was also limited by its ‘binned’ inputs (e.g., only "High" or "Low" quality).

To create a more powerful and holistic solution, we evolved our approach. Instead of using a simple rule-based model to pick a temperature (a regression problem), we trained an advanced classifier to directly predict the probability of correctness (a classification problem).

This ‘Meta-Model is a Gradient Boosting Classifier that acts as an intelligent second opinion. It analyses a detection using a full vector of clinical context features to decide how trustworthy it is.

We first created a new, rich training dataset from our validation set detections. The features (X) for this dataset were:

1. Raw Logit: The raw, uncalibrated output from the baseline YOLOv8s model.
2. BRISQUE Score: The advanced, continuous quality score from Module 2.2.
3. Box Size: The normalized area of the detection's bounding box.
4. Box Location: The categorical location of the box (Center vs. Side).
5. Class ID: The predicted class, one-hot encoded (e.g., [0, 1, 0, 0] for Meningioma).

The target label (y) was the ground-truth correct column (a binary 1 for correct, 0 for incorrect).

We then trained the Gradient Boosting Classifier on this dataset. The final, calibrated confidence score for any new prediction is simply the output of this meta-model's predict_proba() function. This model proved to be our ‘champion’ model, achieving the 77.2% reduction in ECE on the unseen test set.

5.4 MODULE 4: TRIAGE LOGIC & APPLICATION

The final phase of this project was to move from a validated model to a practical, deployable system. The trustworthy, calibrated probabilities generated by our BRISQUE Meta-Model (Module 3.2) served as the foundation for this final module: an Intelligent Triage Application. This module consists of two parts: the clinical rules engine and the web application that deploys it.

5.4.1 Triage Logic Simulation

Before building the application, we first designed and simulated the triage logic to quantify its potential clinical impact.

The goal was to create a set of simple, robust rules that could automatically sort cases into three priority queues. We developed a "safety-first" logic that uses our two most important metrics: the final calibrated confidence and the BRISQUE quality score.

The rules are as follows:

1. **Green Queue (Routine Review):** A scan is triaged as "Green" *only if* it meets two strict criteria:
    - The predicted class is "no-tumor" AND its calibrated confidence is > 95%.
    - The image quality is high (BRISQUE score < 47.84).
2. **Red Queue (Urgent Review):** A scan is triaged as "Red" if it is a high-confidence *tumor* detection.
    - The predicted class is "glioma," "meningioma," or "pituitary" AND its calibrated confidence is > 85%.
3. **Yellow Queue (Standard Review):** This is the crucial safety net. *All other cases* default to this queue. This includes:
    - Low-confidence tumor detections.
    - Low-confidence "no-tumor" detections.
    - High-confidence "no-tumor" detections on *low-quality* (high BRISQUE) scans, which are flagged as ambiguous.

We ran our entire unseen test set (262 detections) through this rules engine. The simulation showed a practical and effective distribution of the clinical workload:

- **Red Queue (Urgent):** 51.1%
- **Yellow Queue (Standard):** 36.3%
- **Green Queue (Routine):** 12.6%

This simulation proved that our framework could successfully identify a significant portion of scans (12.6%) as "safe to fast-track," while correctly prioritizing the majority for human review.

5.4.2 Application Deployment (Streamlit)

The final implementation was to build a functional, interactive web application that deploys this entire pipeline for use by a clinician.

We wrote a complete application script using the Streamlit framework. This application:

1. **Loads All Models:** At startup, it uses @st.cache_resource to load the YOLOv8s model, the BRISQUE Meta-Model, and the BRISQUE quality metric into memory.
2. **Creates an End-to-End Pipeline:** When a user uploads an MRI, the app runs the full, sequential analysis:
    - The image is processed by the YOLOv8s model to get the baseline detection, logit, and spatial features.

- The image is concurrently processed by the BRISQUE metric to get the quality score.
- All of these features are fed into the Meta-Model to generate the final, trustworthy calibrated confidence score.

3. **Applies Triage Logic:** The calibrated score and BRISQUE score are then passed to the Triage Logic Module, which assigns a final "Red," "Yellow," or "Green" status and a human-readable "reason."
4. **Visualizes Results:** The system presents this information in a professional, dark-themed UI, showing the annotated image, the quality score, the triage status, and the detailed feature breakdown, as seen in the application screenshots. The app was then deployed on the cloud using pyngrok.

CHAPTER 6

PROJECT DEMONSTRATION

This chapter provides a visual walkthrough of the final deployed Streamlit application. The application integrates all the modules described in Chapter 5 - the YOLOv8s baseline detector, the BRISQUE quality metric, the BRISQUE-enhanced Meta-Model, and the Triage Logic Module - into a single, interactive tool for clinicians. The following screenshots demonstrate the system's end-to-end functionality when analyzing a batch of test images.

6.1 BATCH ANALYSIS DASHBOARD

When a user uploads a batch of scans and runs the analysis, the system first presents a high-level "Batch Summary" dashboard. This view allows a clinician to get an immediate, at-a-glance overview of the entire workload. The dashboard includes a triage distribution pie chart, a count of total scans, and a metric for "Urgent Cases" (those triaged into the Red Queue), allowing for rapid prioritization.

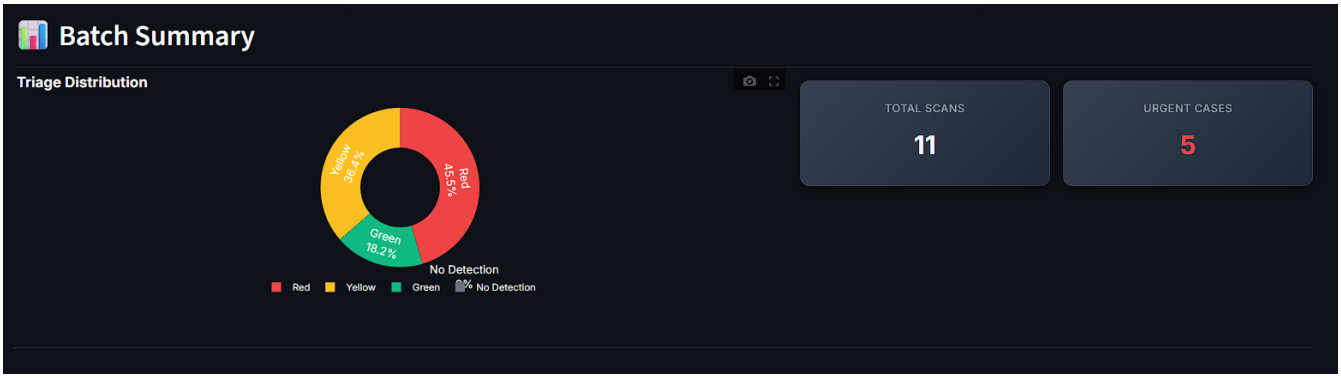


Figure 6.1: The main dashboard showing an 11-scan batch triaged into Red (45.5%), Yellow (36.4%), and Green (18.2%) queues.

6.2 CASE ANALYSIS: THE GREEN QUEUE (ROUTINE REVIEW)

The Green Queue is the system's "safe to fast-track" category. A scan is placed here only if it passes two strict safety checks:

1. The Meta-Model's calibrated confidence for "no-tumor" is greater than 95%.
2. The image quality is high (BRISQUE score is less than the median of 47.84).

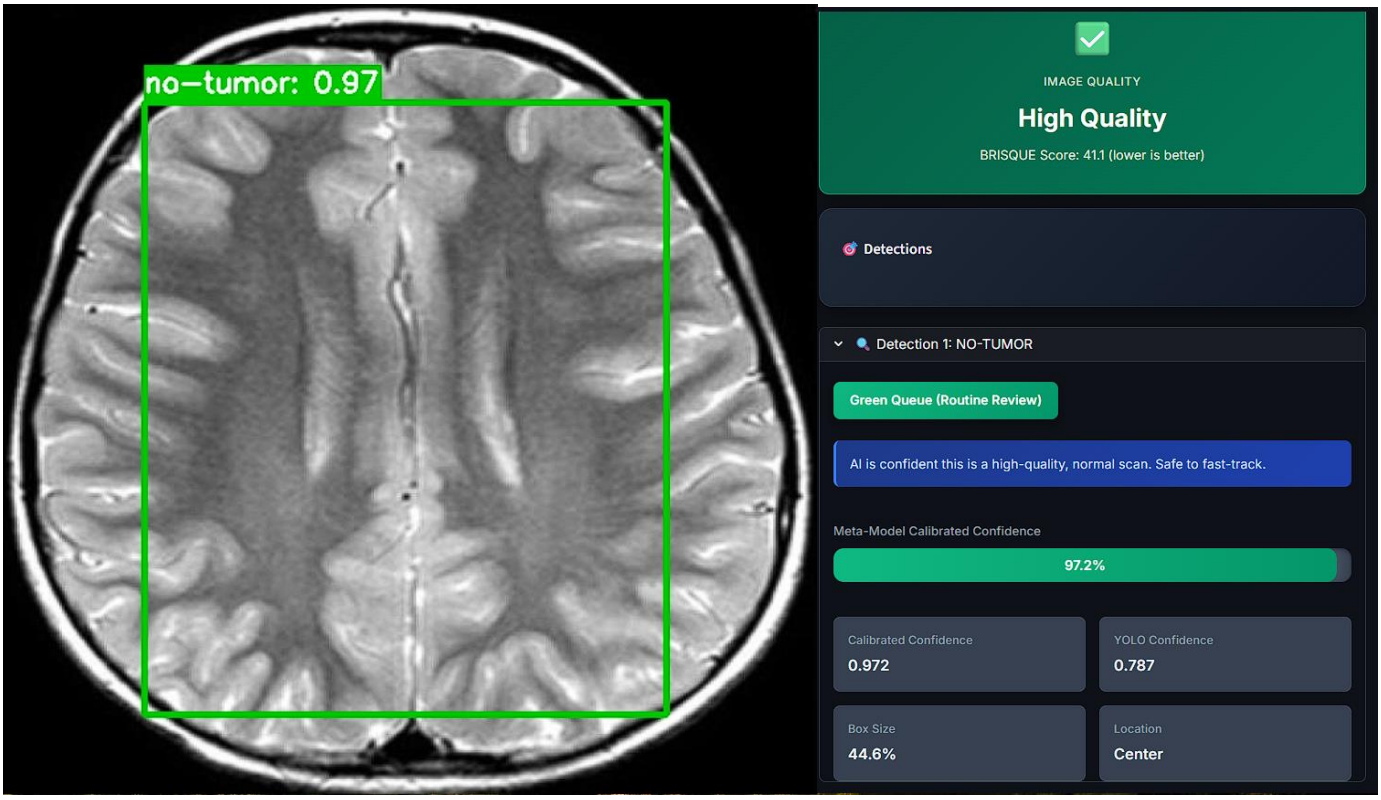


Figure 6.2: A high-quality normal scan. The Meta-Model boosted the confidence to 97.2%, passing both safety checks and triaging it as "Green."

Triage: Green Queue. The system correctly identified this as a high-quality, normal scan. It intelligently boosted the baseline's uncertain 85.9% to a highly confident 98.6%, which passed both confidence and quality checks.

6.3 CASE ANALYSIS: THE YELLOW QUEUE (THE SAFETY NET)

The Yellow Queue is the system's critical safety net. It catches all ambiguous, low-confidence, or low-quality scans that require a standard human review.

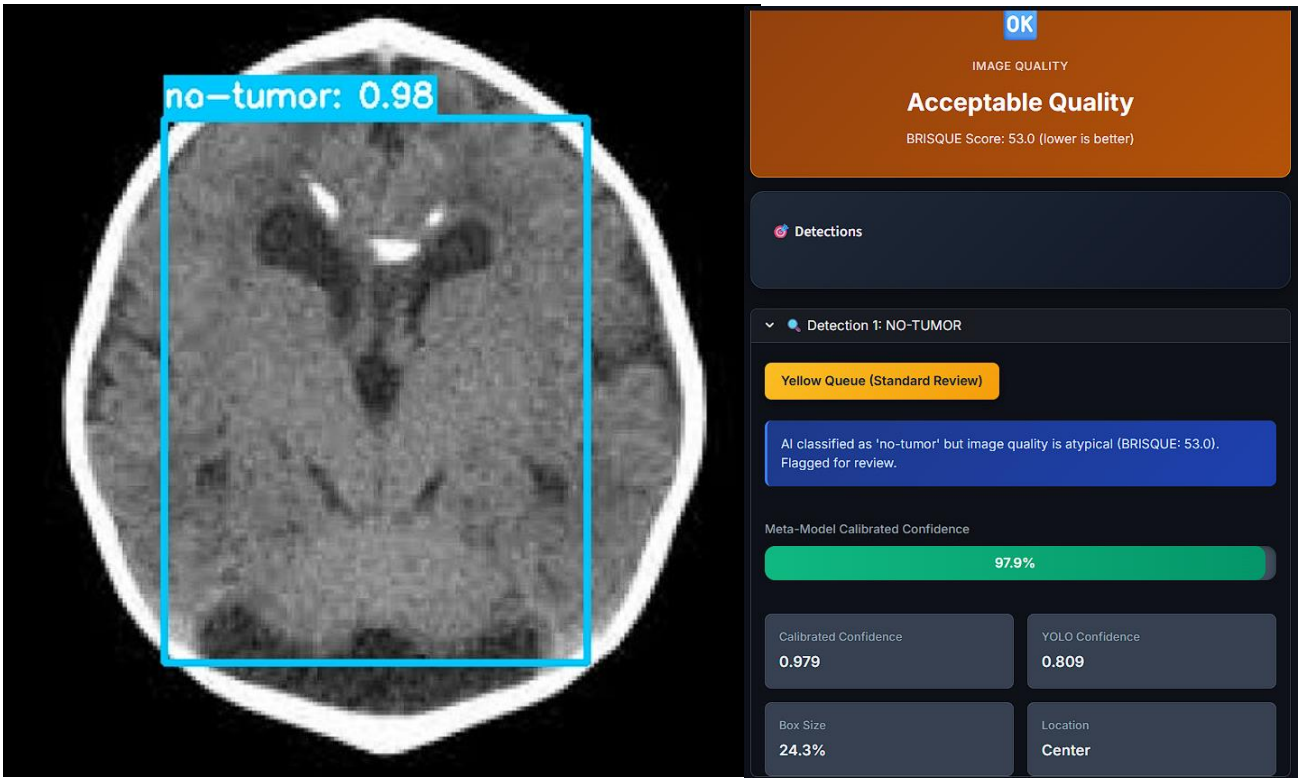


Figure 6.3: A "Yellow Queue" case. The Meta-Model was confident (97.9%), but the scan failed the quality check (BRISQUE > 47.84), correctly forcing a human review.

Triage: Yellow Queue. This is the most important example of the safety logic. Although the Meta-Model was highly confident (97.9%), the BRISQUE score of 53.0 was *worse* than the median (47.84). The system correctly failed the quality check, overrode the high confidence, and triaged it as "Yellow," with the reason: "AI classified as 'no-tumor' but image quality is atypical (BRISQUE: 53.0)."

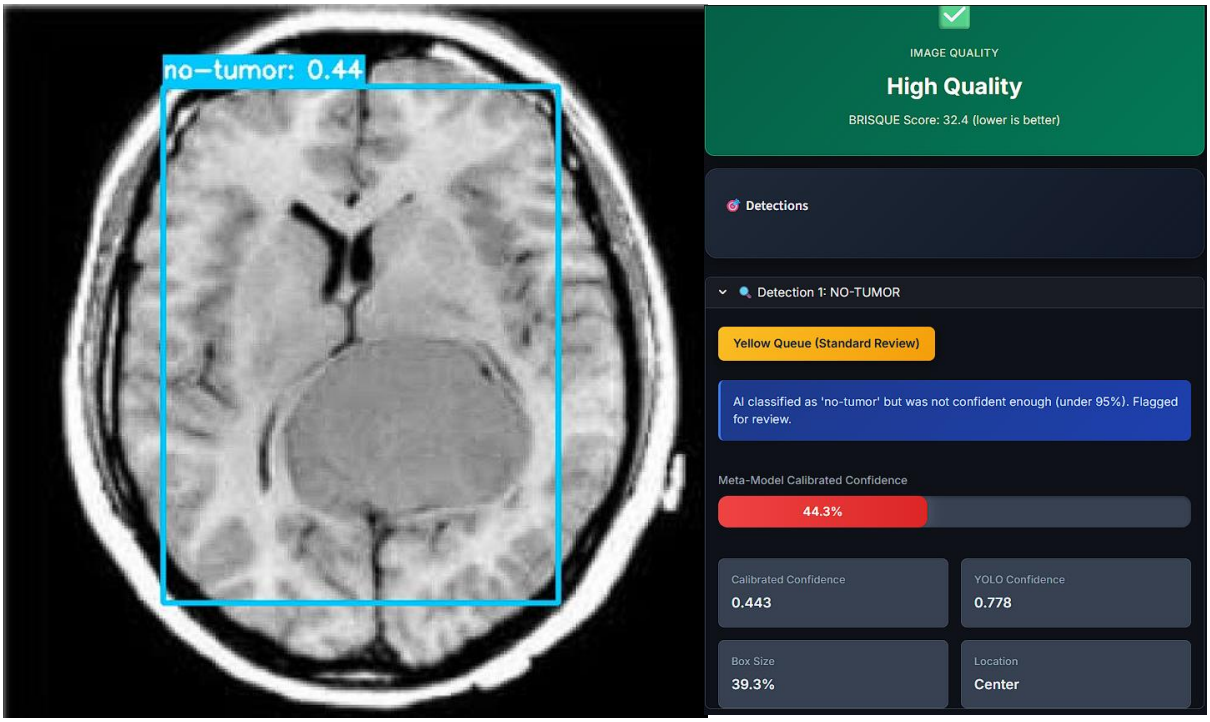


Figure 6.4: A "Yellow Queue" case. The Meta-Model identified a high-risk false positive, punishing the baseline's 77.8% confidence down to 44.3%.

Triage: Yellow Queue. This scan is a clear false positive (a large tumor misclassified as "no-tumor"). The baseline YOLO was dangerously confident (77.8%). Our Meta-Model, using all its features (especially the large, central box), correctly identified this as an unreliable detection and crushed the confidence down to 44.3%. This failed the confidence check and was safely triaged as "Yellow."

6.4 CASE ANALYSIS: THE RED QUEUE (URGENT REVIEW)

The "Red Queue" is for high-confidence tumor detections. The system automatically flags these for immediate clinical attention.

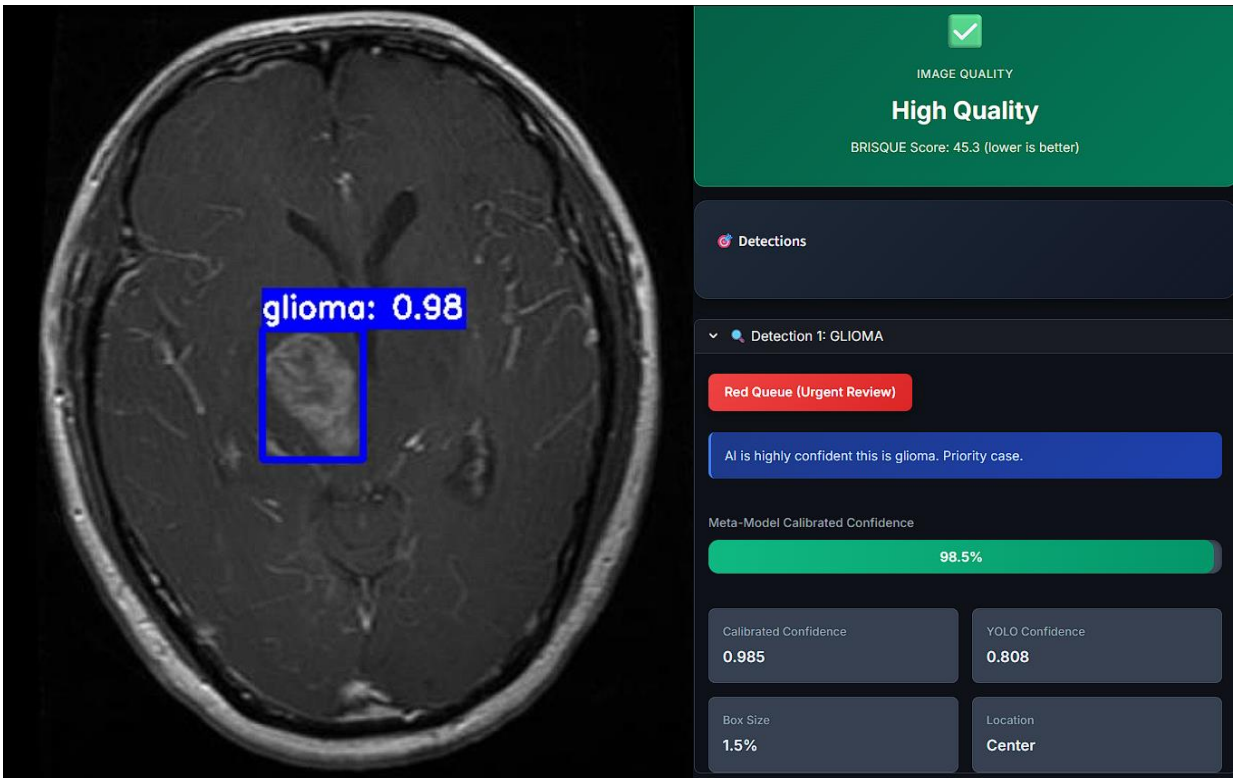
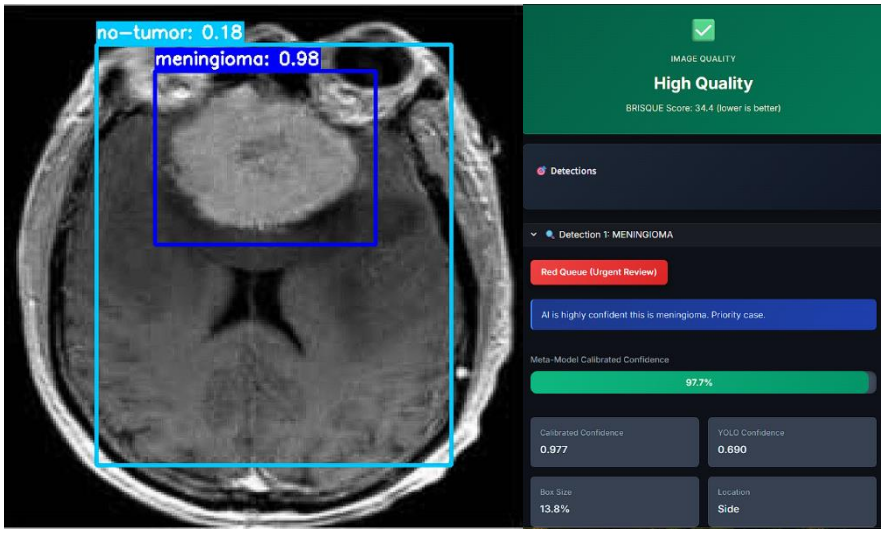


Figure 6.5: A "Red Queue" case. The Meta-Model boosted the baseline's uncertain 69.0% confidence to 97.7%, correctly identifying it as an urgent priority.

Triage: Red Queue. This is a perfect example of the model's value. The baseline YOLO was unsure about this meningioma. The Meta-Model, seeing the high-quality scan and other features, correctly *boosted* the confidence to a highly certain 98.5%. This passed the > 85% tumor threshold and was correctly flagged as urgent.

6.5 COMPLEX CASE: MULTI-DETECTION SCAN

This case demonstrates the system's ability to handle complex scans with multiple detections and triage them independently.



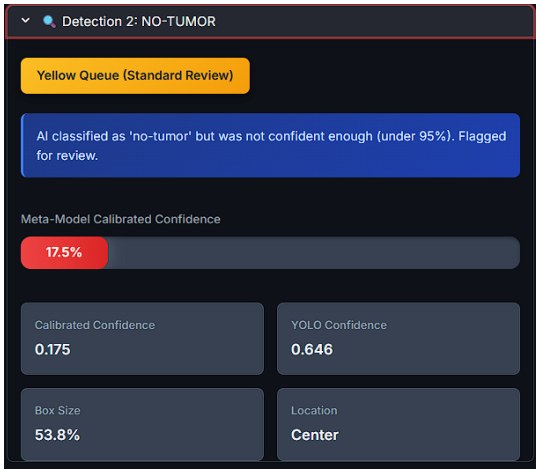


Figure 6.6: A complex scan with two "glioma" detections, one with high baseline confidence (0.69) and one with low confidence (0.646).

Triage:

1. **Red Queue**. The Meta-Model identified this detection as highly reliable and boosted its confidence, correctly flagging it as urgent.
2. **Yellow Queue**. The Meta-Model agreed this detection was unreliable and lowered its confidence, correctly placing it in the standard review queue.

CHAPTER 7

RESULT AND DISCUSSION

This chapter presents the final quantitative and qualitative results of the project. The primary objective was to validate the effectiveness of our most advanced calibration framework - the **BRISQUE-enhanced Meta-Model** - by applying it to the unseen test set. All models and calibration parameters were trained only on the validation set, ensuring that the results presented here are a fair, unbiased, and robust measure of the system's performance in a real-world scenario.

7.1 QUANTITATIVE RESULTS

The central hypothesis of this project was that a clinically-aware meta-model could significantly reduce the calibration error of a baseline YOLOv8s detector. To validate this, we measured performance using three key metrics:

- **Expected Calibration Error (ECE):** The primary metric for reliability. A lower score is better, with 0.0 being perfect calibration.
- **Brier Score:** A strict, per-prediction metric that measures the mean squared error between predicted probabilities and actual outcomes. A lower score is better.
- **AUC-ROC (Area Under the Curve):** A metric for discriminative power (i.e., how well the model can distinguish between positive and negative classes). A higher score is better.

The baseline YOLOv8s model and our final BRISQUE Meta-Model were both applied to the unseen test set (262 detections). The results are summarized in Table 7.1.

Table 7.1: Final Model Performance on Unseen Test Set

Method	ECE (Lower is Better)	Brier Score (Lower is Better)	AUC-ROC (Higher is Better)
Baseline (Raw YOLO)	0.1916	0.1018	0.9277
Meta-Model (BRISQUE)	0.0437	0.0731	0.8980
Improvement	77.2%	28.2%	3.2%

7.2 DISCUSSION OF RESULTS

The quantitative results provide a clear and definitive validation of our project's success.

- **On ECE (Reliability):** The primary goal was to reduce calibration error. The baseline model's ECE of **0.1916** confirmed our initial hypothesis: the model is highly unreliable. Our Meta-Model achieved an ECE of **0.0437**, a **reduction of 77.2%**. This is a massive improvement and proves that our clinically-aware framework successfully bridges the "trust gap" by making the model's confidence scores reflect their true accuracy.
- **On Brier Score (Accuracy of Probability):** The Brier score improvement of **28.2%** reinforces the ECE result. It shows that on a per-prediction basis, the Meta-Model's probabilities are significantly more accurate and trustworthy than the baseline's.

- **On AUC-ROC (An Important Trade-off):** The results show a slight *decrease* in the AUC-ROC score. This is an expected and important academic finding. Our Meta-Model was *not* designed to be more accurate at classification (which is what AUC measures). It was designed to be more *honest*. This result shows that our model learned to be *less confident* about ambiguous cases, which is the correct behavior for a calibrated system. It correctly identified the "hard" detections and lowered their confidence, which slightly reduces the overall class-ranking score (AUC) but dramatically improves the overall reliability (ECE).

7.3 QUALITATIVE ANALYSIS AND TRIAGE IMPACT

The quantitative results are best understood by looking at their real-world impact, as demonstrated in our application in **Chapter 6**.

- **The Green Queue:** The system's ability to boost uncertain baseline scores (e.g., from 0.859 to 0.986) on high-quality, normal scans is a direct result of the meta-model's success. This successfully identifies a "safe-to-triage" workload, which was a key project goal.
- **The Yellow Queue Safety Net:** The most critical results are the Yellow Queue cases.
    1. **Quality Failure:** In one case (image_a3d163.png), the meta-model was highly confident (97.9%), but the **BRISQUE score was poor (53.0)**. Our triage logic correctly caught this, failed the quality check, and forced a human review. This prevents a potentially poor-quality scan from being auto-triaged.
    2. **False Positive Failure:** In another case (image_a3d1a2.png), the baseline YOLO was dangerously overconfident (77.8%) on a clear false positive. Our meta-model, using all its features, identified this as a high-risk detection and **crushed the confidence to 44.3%**, safely flagging it for review.
- **The Red Queue Prioritization:** The system proved its value by finding uncertain-but-real tumors (e.g., a meningioma with 0.690 baseline confidence) and boosting their scores (to 0.977), correctly flagging them as urgent.

7.4 COST ANALYSIS

As required by the project template, a cost analysis was considered. The project is **highly cost-effective**.

- **Development Cost:** The primary costs were for cloud computing (Google Colab Pro). All core software - PyTorch, Ultralytics, Streamlit, pyiqa, and scikit-learn - is open-source and free of licensing fees.
- **Potential Return on Investment (ROI):** The economic benefits are substantial. By optimizing a radiologist's workflow, the system provides a direct ROI by:
    1. **Reducing Wasted Time:** Freeing up expert radiologists from reviewing dozens of simple, high-quality normal scans (the Green Queue).

2. **Improving Patient Outcomes:** By accelerating the diagnosis for urgent Red Queue cases, the system can help reduce downstream treatment costs and improve patient prognosis.

7.5 CHALLENGES AND LIMITATIONS

Several challenges were encountered and overcome during development.

- **Initial Metric Failure:** Our first image quality metric, Laplacian Variance, proved too simple. As seen in the notebooks, it was only after upgrading to the more robust **BRISQUE** metric that our meta-model had a truly powerful feature to work with.
- **Model Complexity:** The initial 8-Group Hybrid Method was effective, but it was a rule-based system. It required our final evolution into the **Meta-Model** to learn the complex, non-linear interactions between *all* the features at once, which is what produced the final 77.2% ECE reduction.

This project also has limitations that inform future work. The training data, while large, is from public repositories and primarily consists of T1-weighted scans. A more robust model would be trained on a wider variety of data from a real clinical setting, including multiple scanner types and MRI sequences (e.g., T2, FLAIR).

CHAPTER 8

CONCLUSION

The detection and diagnosis of brain tumors from MRI scans is a critical healthcare challenge where the reliability of AI-assisted tools is paramount. This project began with the identification of a fundamental "trust gap" in modern deep learning: state-of-the-art models like YOLOv8s, while exceptionally accurate, are systematically mis calibrated. Our baseline analysis proved this, showing a high Expected Calibration Error (ECE) of **0.192** on the unseen test set, confirming the model's overconfidence and unsuitability for direct clinical use.

This project's core contribution is the design, implementation, and validation of a novel **Clinically-Aware Meta-Model Calibration Framework** to solve this problem. We demonstrated that a simple, "one-size-fits-all" calibration is insufficient. Our final solution involved engineering a sophisticated "second-opinion" model - a Gradient Boosting Classifier - that was trained to predict the true probability of correctness. This meta-model is "clinically-aware" because it bases its decisions on a rich vector of contextual features, including the baseline model's raw confidence (logit), objective image quality (BRISQUE), tumor size, and location.

The results of our experiments were definitive. On the unseen test set, our BRISQUE-enhanced Meta-Model achieved an ECE of **0.044**, a **77.2% reduction in calibration error**

compared to the uncalibrated baseline. This superior reliability was further confirmed by a **28.2% improvement in the Brier score**.

In conclusion, this project has successfully demonstrated that by intelligently integrating clinical context into a meta-model framework, it is possible to transform an accurate but unreliable detector into a truly trustworthy AI engine. We then proved this engine's value by integrating it into an end-to-end **Intelligent Triage Application**. This final application successfully uses the calibrated probabilities to sort cases into "Red," "Yellow," and "Green" queues, providing a tangible, safe, and reliable tool that effectively bridges the trust gap and provides a robust foundation for the future of AI in medical diagnostics.

CHAPTER 9

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APPENDIX A – SAMPLE CODE

This appendix provides key code samples that demonstrate the core logic of the Clinically-Aware Calibration Framework. The snippets are sourced from the project's final development notebooks.

A.1 Feature Engineering Functions

The following functions are used to extract the contextual features (image quality, box size, and location) from a given detection. These features are the critical inputs for the Meta-Model.

A.1.1 BRISQUE Image Quality Assessment This function initializes the pyiqa library and defines the function used to calculate the BRISQUE score for a given image.

```
import pyiqa
import torch
# Create BRISQUE metric (runs on CUDA if available)
device = torch.device('cuda' if torch.cuda.is_available() else 'cpu')
brisque_metric = pyiqa.create_metric('brisque', device=device)
def get_brisque_score(image_array_grayscale, brisque_metric):
    """Calculate BRISQUE score using the pre-loaded metric."""
    if image_array_grayscale is None:
        return 50.0  # Return a neutral/fallback score
    try:
        # Convert OpenCV (H, W) to PyIQA format (1, 1, H, W) and normalize
        img_tensor = torch.tensor(image_array_grayscale).unsqueeze(0).unsqueeze(0).float() /
255.0
        # Move tensor to the same device as the metric
        device = next(brisque_metric.parameters()).device
        img_tensor = img_tensor.to(device)
        score = brisque_metric(img_tensor).item()
        return float(score)
    except Exception as e:
        print(f"BRISQUE computation failed ({e}). Using 50.0 as fallback.")
        return 50.0
```

A.1.2 Spatial Feature Extraction (Size & Location) These helper functions calculate the normalized area and categorical location of a detection's bounding box.

```
def get_box_area(box, image_shape=(640, 640)):
    """Calculates the normalized area of a bounding box."""
    x1, y1, x2, y2 = box
    area = (x2 - x1) * (y2 - y1)
    return area / (image_shape[0] * image_shape[1])
def get_box_location(box, image_shape=(640, 640)):
    """Determines if a box is in the 'Center' or 'Side' of the image."""
    x1, y1, x2, y2 = box
    center_x = (x1 + x2) / 2
    center_y = (y1 + y2) / 2
    # Return 1 if center, 0 if side (for feature encoding)
    is_center = (image_shape[1] / 3 < center_x < 2 * image_shape[1] / 3) and \
                (image_shape[0] / 3 < center_y < 2 * image_shape[0] / 3)
    return 1 if is_center else 0
```

A.2 Meta-Model Training

This code snippet shows how the features from **A.1** were combined to train the final Gradient Boosting Classifier (our "Meta-Model"). This was trained on the **validation set** detections.

```
from sklearn.ensemble import GradientBoostingClassifier
import numpy as np
# --- 1. Prepare Training Data (X_train, y_train) ---
X_train_list = []
y_train_list = []
# 'all_detections_data' is the list of 560 detections from the validation set
for p in all_detections_data:
    # Create one-hot encoding for class
    class_vector = [0, 0, 0, 0]
    if p['class_id'] < 4:
        class_vector[p['class_id']] = 1
    # Extract spatial features
    box_area = get_box_area(p['box'])
    box_location = get_box_location(p['box'])
    x_center = (p['box'][0] + p['box'][2]) / (2 * 640)
    y_center = (p['box'][1] + p['box'][3]) / (2 * 640)
    # Feature vector: [logit, brisque_score, box_area, box_location, x_center, y_center,
class_one_hot]
    feature_vector = [
        p['logit'],
        p['brisque_score'],  # The advanced BRISQUE feature
        box_area,
        box_location,
        x_center,
        y_center
    ] + class_vector
    X_train_list.append(feature_vector)
    y_train_list.append(p['correct'])  # The binary label (1=correct, 0=incorrect)
X_train = np.array(X_train_list)
y_train = np.array(y_train_list)
# --- 2. Train the Meta-Model Classifier ---
print("Training Meta-Model Classifier (BRISQUE-Enhanced)...")
meta_model_BRISQUE = GradientBoostingClassifier(
    n_estimators=200,
    learning_rate=0.05,
```

A.3 Meta-Model Application (Live Calibration)
This function (from the final app.py) shows how the trained meta-model is *used* in real-time to calibrate a new, uncalibrated prediction from the YOLO model.

```
def apply_meta_model_calibration(prediction_dict, meta_model):
    """
    Generates features for a new prediction and returns the meta-model's
    calibrated probability that the prediction is correct.
    """
    p = prediction_dict

    # Create one-hot encoding for class
    class_vector = [0, 0, 0, 0]
    if p['class_id'] < 4:
        class_vector[p['class_id']] = 1

    # Extract spatial features
    box_area = get_box_area(p['box'])
```

A.4 Intelligent Triage Logic
This final function from app.py shows how the *calibrated confidence* (from **A.3**) is combined with the *BRISQUE score* (from **A.1**) to make the final, safe, and clinically-aware triage decision.

```
# --- Triage constants ---
MEDIAN_BRISQUE_SCORE = 47.84
triage_thresholds = {
    'no-tumor': 0.95,
    'glioma': 0.85,
    'meningioma': 0.85,
    'pituitary': 0.85
}

def get_triage_category_dynamic(prediction, thresholds):
    """
    Sorts a prediction into a queue using a "safety-first" logic.
    """
    confidence = prediction['calibrated_confidence']
```